

Sid Mathur's Thematic ETF Screener – Daily Report

2026-08-15

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Action Center

Run date: 2026-08-15 **Universe:** 659 ETFs / 26 clusters

Full PC1: LONG since 2026-08-04 (D+8, 1.0%) **Clusters long:** 10/26 **ETFs long:** 341/659 (52%)

Today's events (cluster level)

$\text{spread_buy} = 5\text{dma} - 50\text{dma}$; $\text{spread_sell} = 5\text{dma} - 20\text{dma}$ (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

No new cluster crossovers today.

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **$\text{spread_buy} = 5\text{dma} - 50\text{dma}$, $\text{spread_sell} = 5\text{dma} - 20\text{dma}$** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_4	U.S. Large Cap Value	4.878	1	1	1	1.00
cluster_11	Healthcare	6.420	1	1	1	1.00
cluster_2	U.S. Mid Cap Equity	3.247	1	1	1	1.00
cluster_14	Financials	5.850	1	1	1	0.99
cluster_8	U.S. Dividend Equity	4.277	1	1	1	0.98

Full Sample PC1

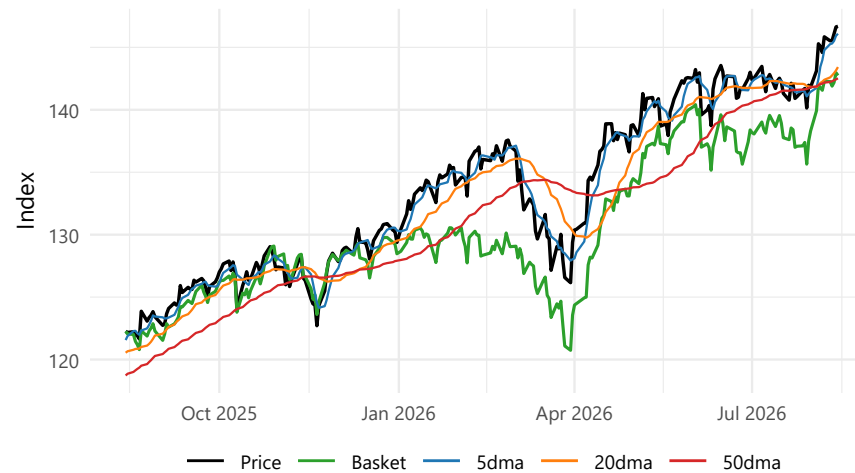


Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: LONG since 2026-08-04 (8 days, 1.0%)

5dma: 146.097 20dma: 143.424 50dma: 142.524

Spread vs 50dma: +3.572 Spread vs 20dma: +2.673

P(buy crossover within 5 days): 100.0% P(sell): 0.0%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
VEA	cluster_1	Developed ex-US Equity	1.000	11	1.8%	8.89%	0.99
VOO	cluster_3	U.S. Large Cap Equity	1.000	7	1.4%	8.02%	0.98
GUNR	cluster_12	Materials	1.000	7	1.9%	5.95%	0.98
EWJ	cluster_22	Japan Equity	1.000	6	2.9%	5.69%	0.97
IGV	cluster_10	Technology	1.000	8	2.0%	3.46%	0.94
XLC	cluster_10	Technology	1.000	8	2.0%	2.94%	0.94
IAU	cluster_17	Gold	1.000	5	0.9%	2.49%	0.95
SLV	cluster_17	Gold	1.000	5	0.9%	1.37%	0.95
AML	cluster_12	Materials	1.000	7	1.9%	1.06%	0.98
URA	cluster_12	Materials	1.000	7	1.9%	0.73%	0.98
GLD	cluster_17	Gold	1.000	5	0.9%	0.08%	0.95
KWEB	cluster_13	China Equity	0.945	9	-0.3%	4.00%	0.92
IEMG	cluster_7	Emerging Markets Equity	0.920	1	-0.2%	4.58%	0.99
QQQ	cluster_6	U.S. Large Cap Growth	0.905	4	1.9%	4.27%	0.98
PDBC	cluster_19	Broad Commodities	0.600	3	-0.2%	5.38%	0.98

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
XLF	cluster_14	Financials	1.00	0.993	1.095	0.96
VTV	cluster_4	U.S. Large Cap Value	1.00	1.000	1.093	0.97
VHT	cluster_11	Healthcare	1.00	1.000	1.088	0.93
IJH	cluster_2	U.S. Mid Cap Equity	1.00	1.000	1.010	0.99
SCHD	cluster_8	U.S. Dividend Equity	1.00	0.975	0.893	0.96
INDA	cluster_20	India Equity	1.00	0.987	0.660	0.91
NOBL	cluster_20	India Equity	1.00	0.987	0.080	0.91
PPLT	cluster_20	India Equity	1.00	0.987	0.029	0.91
BWX	cluster_25	International Fixed Income	0.98	0.931	0.751	0.93

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
1	cluster_1	Developed ex-US Equity	Developed ex-US E...	85	LONG		11	1.8%	1.000	-0.000
2	cluster_2	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	77	FLAT				1.000	+0.000
3	cluster_3	U.S. Large Cap Equity	U.S. Large Cap Eq...	73	LONG		7	1.4%	1.000	+0.001
4	cluster_4	U.S. Large Cap Value	U.S. Large Cap Va...	60	FLAT				1.000	-0.000
10	cluster_10	Technology	Technology	18	LONG		8	2.0%	1.000	+0.000
11	cluster_11	Healthcare	Healthcare	17	FLAT				1.000	+0.000
12	cluster_12	Materials	Materials	17	LONG		7	1.9%	1.000	-0.000
17	cluster_17	Gold	Gold	12	LONG		5	0.9%	1.000	+0.000
22	cluster_22	Japan Equity	Japan Equity	9	LONG		6	2.9%	1.000	+0.000
14	cluster_14	Financials	Financials	14	FLAT				0.993	+0.009
20	cluster_20	India Equity	India Equity	10	FLAT				0.987	-0.012
8	cluster_8	U.S. Dividend Equity	U.S. Dividend Equ...	25	FLAT				0.975	+0.018
13	cluster_13	China Equity	China Equity	15	LONG		9	-0.3%	0.945	-0.054
25	cluster_25	International Fixed Income	International Fix...	6	FLAT				0.931	-0.048
7	cluster_7	Emerging Markets Equity	Emerging Markets ...	25	LONG		1	-0.2%	0.920	+0.127
6	cluster_6	U.S. Large Cap Growth	U.S. Large Cap Gr...	38	LONG		4	1.9%	0.905	+0.091
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	FLAT				0.702	+0.226
19	cluster_19	Broad Commodities	Broad Commodities	11	LONG		3	-0.2%	0.600	+0.142
9	cluster_9	High Yield Bond	High Yield Bond	21	FLAT				0.018	+0.036
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT				-0.053	+0.276
16	cluster_16	Municipal Bond	Municipal Bond	13	FLAT				-0.078	+0.042
5	cluster_5	Aggregate Fixed Income	Aggregate Fixed I...	58	FLAT				-0.649	-0.012
18	cluster_18	Consumer Staples	Consumer Staples	11	FLAT				-0.825	+0.080
15	cluster_15	Real Estate	Real Estate	14	FLAT				-0.973	-0.053
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT				-0.979	+0.011
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT				-0.985	-0.017

Cluster Details

Cluster 1 – Developed ex-US Equity

Members: 85 **Var explained:** 0.813 **Signal:** LONG since 2026-07-30 (11 days, 1.8%)

Signal logic:

- Latest buy_cross: 2026-07-30 – filter **PASS** (252d at date = +20.5%)
- Latest sell_cross: 2026-06-24
- Current 252d return: +19.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-30 > last sell at 2026-06-24)



Category breakdown

category	n_members	abs_weight
International Equity	21	0.2624
Developed ex-US Equity	9	0.1154
International Dividend Equity	6	0.0730
Global ex-US Equity	5	0.0641
Europe Equity	4	0.0507
Alternatives	4	0.0455
Eurozone Equity	3	0.0364
Asia-Pacific Equity	2	0.0238

Real Estate	2	0.0220
Global Equity	2	0.0216
Emerging Markets Fixed Income	2	0.0191
Currency	2	0.0177
Commodity	1	0.0130
Global ex-US Small Cap Equity	1	0.0126
Germany Equity	1	0.0121
France Equity	1	0.0121
United Kingdom Equity	1	0.0120
Italy Equity	1	0.0119
Europe Financials	1	0.0118
Sweden Equity	1	0.0117
Australia Equity	1	0.0117
Netherlands Equity	1	0.0116
Spain Equity	1	0.0113
Canada Equity	1	0.0112
Switzerland Equity	1	0.0111
Austria Equity	1	0.0110
Belgium Equity	1	0.0109
Singapore Equity	1	0.0102
Poland Equity	1	0.0102
South Africa Equity	1	0.0098
Latin America Equity	1	0.0090
New Zealand Equity	1	0.0089
Mexico Equity	1	0.0088
Chile Equity	1	0.0078
Brazil Equity	1	0.0074

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
VEA	0.904	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
3	EFA	0.0130	iShares MSCI EAFE ETF	Developed ex-US Equity	LONG
4	VEA	0.0130	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
51	GSIE	0.0130	Goldman Sachs ActiveBeta International ...	Commodity	LONG
54	ESGD	0.0130	iShares ESG Aware MSCI EAFE ETF	International Equity	LONG
65	IEFA	0.0130	iShares Core MSCI EAFE ETF	Developed ex-US Equity	LONG

67	SPDW	0.0130	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
23	DWM	0.0129	WisdomTree International Equity Fund	International Dividend E...	LONG
36	TLTD	0.0129	FlexShares Morningstar Developed Market...	International Equity	LONG
43	INTF	0.0129	iShares International Equity Factor ETF	International Equity	LONG
61	IXUS	0.0129	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
66	SCHF	0.0129	Schwab International Equity ETF	Developed ex-US Equity	LONG
30	PXF	0.0128	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
40	ACWX	0.0128	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
41	CWI	0.0128	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
44	IQLT	0.0128	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
46	QEFA	0.0128	State Street SPDR MSCI EAFE StrategicFa...	International Equity	LONG
49	VYMI	0.0128	Vanguard International High Dividend Yi...	International Dividend E...	LONG
50	FNDF	0.0128	Schwab Fundamental International Equity...	International Equity	LONG
60	IEUR	0.0128	iShares Core MSCI Europe ETF	Europe Equity	LONG
62	VEU	0.0128	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
63	VGK	0.0128	Vanguard FTSEEuropean ETF	Europe Equity	LONG
64	VXUS	0.0128	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
7	IEV	0.0127	iShares Europe ETF	Europe Equity	LONG
34	EFV	0.0127	iShares MSCI EAFE Value ETF	Developed ex-US Equity	LONG
35	FNDC	0.0127	Schwab Fundamental International Small ...	International Equity	LONG
37	JPIN	0.0127	JPMorgan Diversified Return Internation...	International Equity	LONG
48	RODM	0.0127	Hartford Multifactor Developed Markets ...	International Equity	LONG
58	SCHC	0.0127	Schwab International Small-Cap Equity E...	Developed ex-US Equity	LONG
26	IQDF	0.0126	FlexShares International Quality Divide...	International Equity	LONG
33	SCZ	0.0126	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	LONG
59	VSS	0.0126	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	LONG
9	DLS	0.0125	WisdomTree International SmallCap Fund	International Equity	LONG
21	EZU	0.0125	iShares MSCI Eurozone ETF	Eurozone Equity	LONG
31	EFG	0.0125	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	LONG
45	IVLU	0.0125	iShares MSCI Intl Value Factor ETF	International Equity	LONG
5	FEP	0.0124	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
47	FEZ	0.0124	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	LONG
56	VIGI	0.0124	Vanguard International Dividend Appreci...	International Dividend E...	LONG
76	HFXI	0.0124	NYLI FTSE International Equity Currency...	International Equity	LONG
32	GWX	0.0123	State Street SPDR S&P International Sma...	International Equity	LONG
42	IMTM	0.0123	iShares MSCI Intl Momentum Factor ETF	International Equity	LONG
1	EWG	0.0121	iShares MSCI Germany Index Fund	Germany Equity	LONG
6	VIDI	0.0121	Vident International Equity Strategy ETF	International Equity	LONG
17	EWQ	0.0121	iShares MSCI France Index Fund	France Equity	LONG
22	IDV	0.0121	iShares International Select Dividend E...	International Dividend E...	LONG
82	IPKW	0.0121	Invesco International BuyBack Achievers...	International Equity	LONG
19	EWU	0.0120	iShares MSCI United Kingdom ETF	United Kingdom Equity	LONG
55	HDEF	0.0120	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
24	EPP	0.0119	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	LONG
52	IDLV	0.0119	Invesco S&P International Developed Low...	International Equity	LONG

84	BBAX	0.0119	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
85	EWI	0.0119	iShares MSCI Italy ETF	Italy Equity	FLAT
25	EUFN	0.0118	iShares MSCI Europe Financials ETF	Europe Financials	LONG
14	EWA	0.0117	iShares MSCI Australia Index Fund	Australia Equity	LONG
75	EWD	0.0117	iShares MSCI Sweden ETF	Sweden Equity	LONG
29	DWX	0.0116	State Street SPDR S&P International Div...	International Dividend E...	LONG
38	DBEF	0.0116	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
53	EFAV	0.0116	iShares MSCI EAFE Min Vol Factor ETF	International Equity	LONG
80	EWN	0.0116	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
13	HEZU	0.0115	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	LONG
39	HEFA	0.0115	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
12	PID	0.0113	Invesco International Dividend Achiever...	International Dividend E...	FLAT
16	EWP	0.0113	iShares MSCI Spain ETF	Spain Equity	FLAT
11	IHDG	0.0112	WisdomTree International Hedged Quality...	Alternatives	FLAT
20	EWC	0.0112	iShares MSCI Canada Index Fund	Canada Equity	LONG
57	VNQI	0.0112	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	FLAT
10	HEDJ	0.0111	WisdomTree Europe Hedged Equity Fund	Alternatives	LONG
15	EWL	0.0111	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
27	NFRA	0.0111	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
79	EWO	0.0110	iShares MSCI Austria ETF	Austria Equity	LONG
83	EWK	0.0109	iShares MSCI Belgium ETF	Belgium Equity	LONG
8	RWX	0.0108	State Street SPDR Dow Jones Internation...	Real Estate	LONG
28	IGF	0.0105	iShares Global Infrastructure ETF	Global Equity	FLAT
18	EWS	0.0102	iShares MSCI Singapore ETF	Singapore Equity	LONG
74	EPOL	0.0102	iShares MSCI Poland ETF	Poland Equity	LONG
78	EZA	0.0098	iShares MSCI South Africa Index Fund	South Africa Equity	LONG
68	EBND	0.0096	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
73	FXA	0.0096	Invesco CurrencyShares Australian Dolla...	Currency	LONG
69	EMLC	0.0095	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
72	ILF	0.0090	iShares Latin America 40 ETF	Latin America Equity	FLAT
77	ENZL	0.0089	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
70	EWX	0.0088	iShares MSCI Mexico ETF	Mexico Equity	LONG
81	FXC	0.0082	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
71	ECH	0.0078	iShares MSCI Chile ETF	Chile Equity	LONG
2	EWZ	0.0074	iShares MSCI Brazil ETF	Brazil Equity	FLAT

Cluster 2 – U.S. Mid Cap Equity

Members: 77 Var explained: 0.883 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +39.2%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +22.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
U.S. Small Cap Equity	10	0.1356
Consumer Discretionary	9	0.1067
U.S. Mid Cap Equity	6	0.0814
U.S. Small Cap Value	6	0.0808
U.S. Small Cap Growth	6	0.0793
Industrials	4	0.0471
U.S. Mid Cap Value	3	0.0407
U.S. Mid Cap Growth	3	0.0397
Leveraged / Inverse	2	0.0259
Dividend Equity	2	0.0257
U.S. Small Cap Low Volatility	2	0.0245
U.S. Small Cap Fundamental Equity	1	0.0137
U.S. Small Cap ESG Equity	1	0.0137
Unknown	1	0.0137
U.S. Mid/Small Cap Equity	1	0.0136
U.S. Small/Mid Cap Fundamental Equity	1	0.0136

Mid Cap Value Equities	1	0.0136
U.S. Small Cap Multifactor Equity	1	0.0136
Small Cap Value Equities	1	0.0134
U.S. Small/Mid Cap Equity	1	0.0133
U.S. Extended Market Equity	1	0.0133
U.S. ESG Equity	1	0.0133
U.S. Mid Cap Dividend Equity	1	0.0132
Growth Equity	1	0.0131
U.S. Mid Cap Quality Equity	1	0.0131
Value Equity	1	0.0131
U.S. Equity	1	0.0131
U.S. Small Cap Dividend Equity	1	0.0131
U.S. Small Cap Momentum Equity	1	0.0128
U.S. Small Cap Industrials	1	0.0128
U.S. Micro Cap Equity	1	0.0126
U.S. Small Cap Consumer Discretionary	1	0.0124
U.S. Dividend Equity	1	0.0122
U.S. Mid Cap Low Volatility	1	0.0117
Real Estate	1	0.0106

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.012

ticker	rep_wt	name	category	etf_signal
IJH	1.01	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	NUSC	0.0137	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
19	MDY	0.0137	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
27	IVOO	0.0137	Vanguard S&P Mid-Cap 400 ETF		FLAT
28	VI00	0.0137	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
31	IJR	0.0137	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
34	IJH	0.0137	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
35	SPMD	0.0137	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
36	SPSM	0.0137	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
37	VB	0.0137	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
51	FNDA	0.0137	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
63	VBR	0.0137	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
12	FNX	0.0136	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
13	FYX	0.0136	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT

16	PRFZ	0.0136	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
17	SMLF	0.0136	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	LONG
38	SCHA	0.0136	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
48	EZM	0.0136	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
59	MDYV	0.0136	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
5	IWM	0.0135	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	LONG
9	JHMM	0.0135	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
11	SCHM	0.0135	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
20	IJT	0.0135	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
22	VIOG	0.0135	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
25	SLYG	0.0135	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
29	VTWO	0.0135	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	LONG
54	IWN	0.0135	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
56	IJJ	0.0135	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
58	IVOV	0.0135	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
62	VTWV	0.0135	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
47	EES	0.0134	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
49	PSC	0.0134	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
57	IJS	0.0134	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
60	SLYV	0.0134	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
61	VIOV	0.0134	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
69	RWK	0.0134	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
76	SDVY	0.0134	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
33	VXF	0.0133	Vanguard Extended Market ETF	U.S. Extended Market Equ...	LONG
64	ETHO	0.0133	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
65	RWJ	0.0133	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
77	ISMD	0.0133	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
21	IJK	0.0132	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
23	IVOG	0.0132	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
24	MDYG	0.0132	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
46	DON	0.0132	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
32	ISCG	0.0131	iShares Morningstar Small-Cap Growth ETF	Growth Equity	LONG
45	DES	0.0131	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
67	XMHQ	0.0131	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
68	RFV	0.0131	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
70	OMFS	0.0131	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
75	FYT	0.0131	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
18	IWO	0.0129	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
26	VTWG	0.0129	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
71	RZV	0.0129	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
30	VBK	0.0128	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG
39	FXD	0.0128	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	FLAT
66	PSCI	0.0128	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
74	XSMO	0.0128	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	LONG
14	IWC	0.0126	iShares Microcap ETF	U.S. Micro Cap Equity	LONG

43	RSPD	0.0126	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	LONG
55	SMMV	0.0125	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
41	FTXR	0.0124	First Trust Nasdaq Transportation ETF	Industrials	LONG
73	PSCD	0.0124	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
10	REGL	0.0123	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
44	SMDV	0.0122	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
50	XTN	0.0122	State Street SPDR S&P Transportation ETF	Industrials	FLAT
2	XRT	0.0120	State Street SPDR S&P Retail ETF	Consumer Discretionary	LONG
6	IYC	0.0120	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
53	XSLV	0.0120	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
8	FDIS	0.0119	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG
7	VCR	0.0118	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG
42	IYT	0.0118	iShares U.S. Transportation ETF	Industrials	LONG
40	PEJ	0.0117	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
52	XMLV	0.0117	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	FLAT
1	XLY	0.0114	State Street Consumer Discretionary Sel...	Consumer Discretionary	LONG
3	JETS	0.0107	U.S. Global Jets ETF	Industrials	FLAT
4	REM	0.0106	iShares Mortgage Real Estate ETF	Real Estate	LONG
72	RTH	0.0105	VanEck Retail ETF	Consumer Discretionary	LONG

Cluster 3 – U.S. Large Cap Equity

Members: 73 **Var explained:** 0.872 **Signal:** LONG since 2026-08-05 (7 days, 1.4%)

Signal logic:

- Latest buy_cross: 2026-08-05 – filter **PASS** (252d at date = +21.4%)
- Latest sell_cross: 2026-07-17
- Current 252d return: +21.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-05 > last sell at 2026-07-17)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1308
U.S. Total Market Equity	7	0.1021
Global Equity	7	0.1002
Technology	5	0.0663
Asset Allocation	4	0.0532
Industrials	4	0.0453
U.S. Equity	3	0.0426
U.S. Multifactor Equity	3	0.0425
U.S. Mega Cap Equity	3	0.0424
U.S. Quality / Factor Equity	3	0.0421
U.S. Mid Cap Growth	3	0.0419
U.S. Momentum Equity	3	0.0382
U.S. ESG Equity	2	0.0290
Momentum Equity	2	0.0264
Convertible Bond	2	0.0251
Commodity	1	0.0145
Large Cap Blend Equities	1	0.0144
Large Cap Value Equities	1	0.0143
Dividend Equity	1	0.0141
U.S. Equal Weight Equity	1	0.0138
U.S. Dividend Equity	1	0.0138
Growth Equity	1	0.0136

IPO Equity	1	0.0131
U.S. Mid Cap Momentum Equity	1	0.0130
Alternatives	1	0.0128
U.S. Small Cap Momentum Equity	1	0.0124
International IPO Equity	1	0.0113
Israel Equity	1	0.0107

Replication basket – VOO (cor=0.98) basket lead at lag 1: -0.008

ticker	rep_wt	name	category	etf_signal
VOO	0.996	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
43	IWV	0.0146	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
44	IYY	0.0146	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
50	IWB	0.0146	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG
55	VTHR	0.0146	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
58	VONE	0.0146	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
62	ITOT	0.0146	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG
64	SCHB	0.0146	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
65	SCHX	0.0146	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
66	SPTM	0.0146	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
68	VTI	0.0146	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
3	SPY	0.0145	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
5	ACWI	0.0145	iShares MSCI ACWI ETF	Global Equity	LONG
10	URTH	0.0145	iShares MSCI World ETF	Global Equity	LONG
13	VT	0.0145	Vanguard Total World Stock Index ETF	Global Equity	LONG
30	TILT	0.0145	FlexShares Morningstar US Market Factors...	U.S. Multifactor Equity	FLAT
39	JHML	0.0145	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
42	SUSA	0.0145	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG
47	SPYX	0.0145	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
49	ESGU	0.0145	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
56	GSLC	0.0145	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
60	VV	0.0145	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
61	ILCB	0.0145	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
63	IVV	0.0145	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
67	VOO	0.0145	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
11	CRBN	0.0144	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
12	SPGM	0.0144	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG

41	DSI	0.0144	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG
45	LRGF	0.0144	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
51	IWL	0.0144	iShares Russell Top 200 ETF	U.S. Equity	LONG
59	MGC	0.0144	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
6	AOA	0.0143	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
57	EPS	0.0143	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
46	OEF	0.0142	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
52	QUAL	0.0142	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	FLAT
29	IMCG	0.0141	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
37	QDF	0.0141	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
28	VOT	0.0140	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
38	CATH	0.0140	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
9	AOR	0.0139	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
25	IWP	0.0139	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
36	IOO	0.0139	iShares Global 100 ETF	Global Equity	LONG
48	XLG	0.0139	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	LONG
53	QUS	0.0139	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT
54	SPHQ	0.0139	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
16	QQEW	0.0138	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
17	XT	0.0138	iShares Future Exponential Technologies...	Technology	LONG
19	FTC	0.0138	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
35	SPHB	0.0138	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
40	DGRW	0.0138	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
23	RPG	0.0136	Invesco S&P 500 Pure Growth ETF	Growth Equity	FLAT
73	OMFL	0.0136	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
2	ROBO	0.0134	ROBO Global Robotics and Automation Ind...	Technology	LONG
20	PDP	0.0134	Invesco Dorsey Wright Momentum ETF	Momentum Equity	FLAT
70	VFMO	0.0133	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
1	BOTZ	0.0131	Global X Robotics & Artificial Intellig...	Technology	LONG
21	FPX	0.0131	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
8	AOM	0.0130	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
14	FV	0.0130	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
24	XMMO	0.0130	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	FLAT
27	MTUM	0.0129	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	FLAT
4	QAI	0.0128	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
22	CWB	0.0128	State Street SPDR Bloomberg Convertible...	Convertible Bond	LONG
31	DWAS	0.0124	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	LONG
26	ICVT	0.0123	iShares Convertible Bond ETF	Convertible Bond	FLAT
72	PRN	0.0122	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
15	PRNT	0.0121	3D Printing (The) ETF	Technology	LONG
69	QMOM	0.0120	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	LONG
7	AOK	0.0119	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
18	FPXI	0.0113	First Trust International Equity Opport...	International IPO Equity	FLAT
32	PPA	0.0113	Invesco Aerospace & Defense ETF	Industrials	LONG
34	XAR	0.0113	State Street SPDR S&P Aerospace & Defen...	Industrials	LONG

71	EIS	0.0107	iShares MSCI Israel ETF	Israel Equity	LONG
33	ITA	0.0105	iShares U.S. Aerospace & Defense ETF	Industrials	LONG

Cluster 4 – U.S. Large Cap Value

Members: 60 **Var explained:** 0.890 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +26.6%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +20.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Value	10	0.1728
U.S. Dividend Equity	8	0.1359
Industrials	7	0.1143
Materials	4	0.0627

U.S. Mid Cap Equity	3	0.0515
U.S. Multifactor Equity	3	0.0512
U.S. Value Factor Equity	3	0.0502
Utilities	3	0.0468
U.S. Fundamental Equity	2	0.0348
U.S. Equal Weight Equity	2	0.0347
U.S. Mid Cap Value	2	0.0346
U.S. Equity	2	0.0343
Consumer Discretionary	2	0.0272
Large Cap Blend Equities	1	0.0173
U.S. Size Factor Equity	1	0.0172
U.S. Revenue Weighted Equity	1	0.0171
Value Equity	1	0.0170
U.S. Mega Cap Value	1	0.0169
U.S. Buyback Equity	1	0.0168
U.S. Large Cap Equity	1	0.0167
U.S. Moat Equity	1	0.0161
Communication Services	1	0.0137

Replication basket – VTV (cor=0.97) basket lead at lag 1: -0.003

ticker	rep_wt	name	category	etf_signal
VTV	1.093	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
3	RSP	0.0175	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
26	IWD	0.0175	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
36	VONV	0.0175	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
15	PRF	0.0174	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
18	CFA	0.0174	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
22	FNDX	0.0174	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
29	JPUS	0.0174	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
42	SCHV	0.0174	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
48	IWS	0.0174	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
6	FEX	0.0173	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
39	ILCV	0.0173	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	FLAT
41	IUSV	0.0173	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	FLAT
59	JPME	0.0173	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
23	EQAL	0.0172	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT

27	IWR	0.0172	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
28	IVE	0.0172	iShares S&P 500 Value ETF	U.S. Large Cap Value	FLAT
30	SIZE	0.0172	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	FLAT
34	VOOV	0.0172	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	FLAT
40	IMCB	0.0172	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
43	SPYV	0.0172	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	FLAT
45	VTV	0.0172	Vanguard Value ETF	U.S. Large Cap Value	LONG
49	VOE	0.0172	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
52	DTD	0.0172	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
16	RWL	0.0171	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
20	NULV	0.0171	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	FLAT
44	VO	0.0171	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
54	DGRO	0.0171	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
60	JVAL	0.0171	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
12	IYJ	0.0170	iShares U.S. Industrials ETF	Industrials	FLAT
17	AIVL	0.0170	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
24	IWX	0.0170	iShares Russell Top 200 Value ETF	Value Equity	LONG
25	ONEV	0.0170	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
38	VYM	0.0170	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	FLAT
50	PFM	0.0170	Invesco Dividend Achievers ETF	U.S. Dividend Equity	FLAT
51	DLN	0.0170	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
55	VIG	0.0170	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
37	MGV	0.0169	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	FLAT
5	PKW	0.0168	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
10	RDVY	0.0168	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
21	FDVV	0.0168	Fidelity High Dividend ETF	U.S. Dividend Equity	FLAT
4	FXR	0.0167	First Trust Industrials AlphaDEX	Industrials	FLAT
53	DIA	0.0167	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	FLAT
33	VIS	0.0166	Vanguard Industrials ETF	Industrials	LONG
35	FIDU	0.0166	Fidelity MSCI Industrials Index ETF	Industrials	LONG
1	XLI	0.0165	State Street Industrial Select Sector S...	Industrials	FLAT
19	CFO	0.0164	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
56	PAVE	0.0162	Global X U.S. Infrastructure Developmen...	Industrials	LONG
11	MOAT	0.0161	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
31	VLUE	0.0161	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
9	FIW	0.0160	First Trust Water ETF	Utilities	FLAT
32	VAW	0.0160	Vanguard Materials ETF	Materials	LONG
2	XLB	0.0158	State Street Materials Select Sector SP...	Materials	LONG
7	PHO	0.0158	Invesco Water Resources ETF	Utilities	FLAT
14	RSPM	0.0156	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
13	IYM	0.0152	iShares U.S. Basic Materials ETF	Materials	LONG
8	CGW	0.0150	Invesco S&P Global Water Index ETF	Utilities	FLAT
58	PKB	0.0148	Invesco Building & Construction ETF	Industrials	FLAT
47	XHB	0.0142	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
57	IYZ	0.0137	iShares U.S. Telecommunications ETF	Communication Services	LONG

46 ITB 0.0130 iShares U.S. Home Construction ETF Consumer Discretionary LONG

Cluster 5 – Aggregate Fixed Income

Members: 58 Var explained: 0.831 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +4.5%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +0.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3284
Investment Grade Corporate Bond	13	0.2303
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0891

Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0306
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.006

ticker	rep_wt	name	category	etf_signal
BND	1.007	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	FLAT
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	FLAT
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	FLAT
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	FLAT
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
46	SCHR	0.0183	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	FLAT
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	FLAT
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
27	USIG	0.0181	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	FLAT
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
50	MBB	0.0181	iShares MBS ETF	Mortgage-Backed Fixed In...	FLAT
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	FLAT
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	FLAT
52	SPMB	0.0180	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	FLAT
3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	FLAT
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	FLAT
10	CORP	0.0178	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	FLAT
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT

8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	FLAT
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	FLAT
44	SPTL	0.0172	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
16	SPLB	0.0171	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	FLAT
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT
39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	FLAT
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	FLAT
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	FLAT
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	FLAT
33	BSCQ	0.0155	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	LONG
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
14	IBDR	0.0151	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
38	CMBS	0.0135	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
32	LDUR	0.0121	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 6 – U.S. Large Cap Growth

Members: 38 **Var explained:** 0.893 **Signal:** LONG since 2026-08-10 (4 days, 1.9%)

Signal logic:

- Latest buy_cross: 2026-08-10 – filter **PASS** (252d at date = +36.3%)
- Latest sell_cross: 2026-07-16
- Current 252d return: +35.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-10 > last sell at 2026-07-16)



Category breakdown

category	n_members	abs_weight
Technology	19	0.4990
U.S. Large Cap Growth	11	0.2988
Growth Equity	2	0.0536
U.S. Equity	1	0.0273
U.S. Mega Cap Growth	1	0.0270
Options Income	1	0.0246
U.S. Small Cap Technology	1	0.0243
Trend-Following Equity	1	0.0242
U.S. Trend-Following Equity	1	0.0211

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.030

ticker	rep_wt	name	category	etf_signal
QQQ	1.032	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
16	IGM	0.0276	iShares Expanded Tech Sector ETF	Technology	LONG
5	QQQ	0.0275	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
18	IYW	0.0275	iShares U.S. Technology ETF	Technology	LONG
24	VGT	0.0275	Vanguard Information Tech ETF	Technology	LONG
26	FTEC	0.0275	Fidelity MSCI Information Technology In...	Technology	LONG
29	ILCG	0.0274	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
1	XLK	0.0273	State Street Technology Select Sector S...	Technology	LONG
17	IXN	0.0273	iShares Global Tech ETF	Technology	LONG
20	ONEQ	0.0273	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
22	IWF	0.0272	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
23	IVW	0.0272	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
25	VOOG	0.0272	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
27	VONG	0.0272	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
30	IUSG	0.0272	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
32	SPYG	0.0272	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
12	XNTK	0.0271	State Street SPDR NYSE Technology ETF	Technology	LONG
33	VUG	0.0271	Vanguard Growth ETF	U.S. Large Cap Growth	LONG
19	NULG	0.0270	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
21	IWY	0.0270	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
28	MGK	0.0270	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
7	QTEC	0.0269	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
31	SCHG	0.0268	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
4	AIQ	0.0266	Global X Artificial Intelligence & Tech...	Technology	LONG
11	FXL	0.0266	First Trust Technology AlphaDEX	Technology	LONG
15	PWB	0.0266	Invesco Large Cap Growth ETF	Growth Equity	LONG
3	SMH	0.0259	VanEck Semiconductor ETF	Technology	FLAT
9	TDIV	0.0259	First Trust NASDAQ Technology Dividend ...	Technology	LONG
2	SOXX	0.0256	iShares PHLX SOX Semiconductor Sector I...	Technology	FLAT
38	QTUM	0.0255	Defiance Quantum ETF	Technology	LONG
10	XSD	0.0250	State Street SPDR S&P Semiconductor ETF	Technology	FLAT
36	FTXL	0.0250	First Trust Nasdaq Semiconductor ETF	Technology	FLAT
8	PSI	0.0249	Invesco Semiconductors ETF	Technology	FLAT
6	NXTG	0.0248	First Trust Indxx NextG ETF	Technology	LONG
14	QYLD	0.0246	Global X NASDAQ 100 Covered Call ETF	Options Income	LONG
37	PTF	0.0244	Invesco Dorsey Wright Technology Moment...	Technology	LONG
35	PSCT	0.0243	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	FLAT
13	PTNQ	0.0242	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
34	PTLC	0.0211	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG

Cluster 7 – Emerging Markets Equity

Members: 25 **Var explained:** 0.870 **Signal:** LONG since 2026-08-13 (1 days, -0.2%)

Signal logic:

- Latest buy_cross: 2026-08-13 – filter **PASS** (252d at date = +28.6%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +27.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-13 > last sell at 2026-07-07)

**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7729
Asia Equity	2	0.0815
Asia ex-Japan Equity	1	0.0419
Taiwan Equity	1	0.0362
South Korea Equity	1	0.0339
Global Dividend Equity	1	0.0335

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.034

ticker	rep_wt	name	category	etf_signal
IEMG	0.917	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0426	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	LONG
16	GEM	0.0423	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	LONG
20	ESGE	0.0423	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	LONG
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	FLAT
5	AAXJ	0.0419	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	LONG
15	EMGF	0.0419	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	LONG
11	EEMA	0.0415	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	LONG
14	GMF	0.0412	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	LONG
17	FNDE	0.0409	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
12	PXH	0.0408	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	LONG
19	EEMV	0.0405	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	LONG
10	AIA	0.0403	iShares Asia 50 ETF	Asia Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	LONG
7	DEM	0.0397	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0392	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0389	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	LONG
13	DVYE	0.0363	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	LONG
9	EWI	0.0362	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
25	FEMS	0.0349	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	LONG
1	EWY	0.0339	iShares MSCI South Korea ETF	South Korea Equity	FLAT
24	SDIV	0.0335	Global X SuperDividend ETF	Global Dividend Equity	LONG

Cluster 8 – U.S. Dividend Equity

Members: 25 **Var explained:** 0.869 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-20 – filter **PASS** (252d at date = +20.8%)

- Latest sell_cross: 2026-06-24
- Current 252d return: +18.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-24 more recent than last effective buy 2026-04-20)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	13	0.5283
U.S. Low Volatility Equity	4	0.1562
Value Equity	3	0.1200
Dividend Equity	2	0.0792
U.S. Defensive Equity	1	0.0404
U.S. Quality / Factor Equity	1	0.0390
Global Equity	1	0.0368

Replication basket – SCHD (cor=0.96) basket lead at lag 1: 0.005

ticker	rep_wt	name	category	etf_signal
SCHD	0.893	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

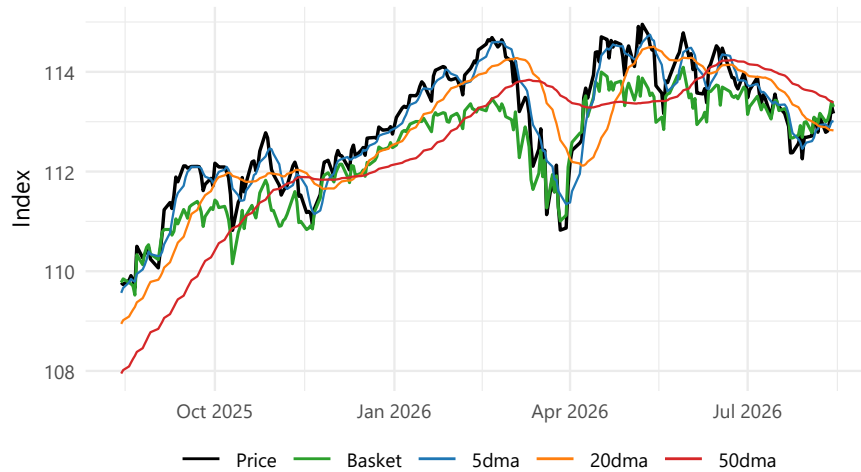
	ticker	PC_wt	name	category	etf_signal
7	SDY	0.0418	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
15	DVY	0.0417	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
2	FVD	0.0415	First Trust VL Dividend	U.S. Dividend Equity	LONG
10	SCHD	0.0413	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
14	SDOG	0.0413	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	FLAT
11	FTA	0.0412	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
20	SPYD	0.0412	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	FLAT
5	DHS	0.0411	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
19	ONEY	0.0411	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	FLAT
18	SPHD	0.0409	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
6	CDC	0.0404	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
4	FDL	0.0402	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
25	LGLV	0.0401	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
8	LVHD	0.0399	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	FLAT
3	PWV	0.0398	Invesco Large Cap Value ETF	Value Equity	LONG
9	HDV	0.0394	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
16	RDIV	0.0394	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	FLAT
12	PEY	0.0393	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
17	RPV	0.0391	Invesco S&P 500 Pure Value ETF	Value Equity	FLAT
13	DIV	0.0390	Global X Super Dividend ETF	U.S. Dividend Equity	FLAT
21	FTCS	0.0390	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
22	OUSA	0.0383	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
23	SPLV	0.0381	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	FLAT
24	USMV	0.0381	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	FLAT
1	ACWV	0.0368	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG

Cluster 9 – High Yield Bond

Members: 21 **Var explained:** 0.764 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +2.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5158
Preferred Stock	7	0.3056
Emerging Markets Fixed Income	4	0.1786

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.046

ticker	rep_wt	name	category	etf_signal
HYG	0.917	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	JNK	0.0527	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT
1	HYG	0.0526	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	FLAT
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	FLAT
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	FLAT

14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	FLAT
17	FALN	0.0517	iShares Fallen Angels USD Bond ETF	High Yield Bond	FLAT
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
6	HYS	0.0512	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
9	PFF	0.0479	iShares Preferred and Income Securities...	Preferred Stock	LONG
3	HYLS	0.0472	First Trust Tactical High Yield ETF	High Yield Bond	LONG
2	EMB	0.0468	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0467	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
11	PFXF	0.0458	VanEck Preferred Securities ex Financia...	Preferred Stock	LONG
20	PCY	0.0458	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0433	First Trust Preferred Securities and In...	Preferred Stock	FLAT
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	FLAT
8	VRP	0.0368	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 10 – Technology

Members: 18 **Var explained:** 0.803 **Signal:** LONG since 2026-08-04 (8 days, 2.0%)

Signal logic:

- Latest buy_cross: 2026-08-04 – filter **PASS** (252d at date = +6.2%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +3.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-04 > last sell at 2026-07-23)



Category breakdown

category	n_members	abs_weight
Technology	10	0.5588
Communication Services	3	0.1637
Financials	2	0.1126
Consumer Discretionary	2	0.1080
IPO Equity	1	0.0569

Replication basket – XLC+IGV (cor=0.94) basket lead at lag 1: 0.033

ticker	rep_wt	name	category	etf_signal
XLC	0.508	State Street Communication Services S...	Communication Services	LONG
IGV	0.597	iShares Expanded Tech-Software Sector...	Technology	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
14	FDN	0.0597	First Trust DJ Internet Index Fund	Technology	LONG
13	PNQI	0.0592	Invesco Nasdaq Internet ETF	Technology	FLAT

7	ARKW	0.0583	ARK Next Generation Internet ETF	Technology	FLAT
16	FINX	0.0580	Global X FinTech ETF	Financials	FLAT
3	SKYY	0.0578	First Trust Cloud Computing ETF	Technology	LONG
6	ARKK	0.0571	ARK Innovation ETF	Technology	LONG
12	IPO	0.0569	Renaissance IPO ETF	IPO Equity	LONG
10	IBUY	0.0567	Amplify Online Retail ETF	Consumer Discretionary	FLAT
2	IGV	0.0562	iShares Expanded Tech-Software Sector E...	Technology	FLAT
18	FCOM	0.0553	Fidelity MSCI Communication Services In...	Communication Services	LONG
17	VOX	0.0552	Vanguard Communication Services ETF	Communication Services	LONG
15	IPAY	0.0546	Amplify Digital Payments ETF	Financials	FLAT
8	ARKQ	0.0542	ARK Autonomous Technology & Robotics ETF	Technology	LONG
11	CIBR	0.0536	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
1	XLC	0.0532	State Street Communication Services Sel...	Communication Services	LONG
5	HACK	0.0528	Amplify Cybersecurity ETF	Technology	FLAT
9	ESPO	0.0512	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
4	BLOK	0.0499	Amplify Blockchain Technology ETF	Technology	FLAT

Cluster 11 – Healthcare

Members: 17 **Var explained:** 0.755 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +31.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9418
U.S. Small Cap Healthcare	1	0.0582

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.003

ticker	rep_wt	name	category	etf_signal
VHT	1.088	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0656	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0655	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0648	First Trust Health Care AlphaDEX	Healthcare	FLAT
14	RSPH	0.0642	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0640	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0625	iShares Global Healthcare ETF	Healthcare	FLAT

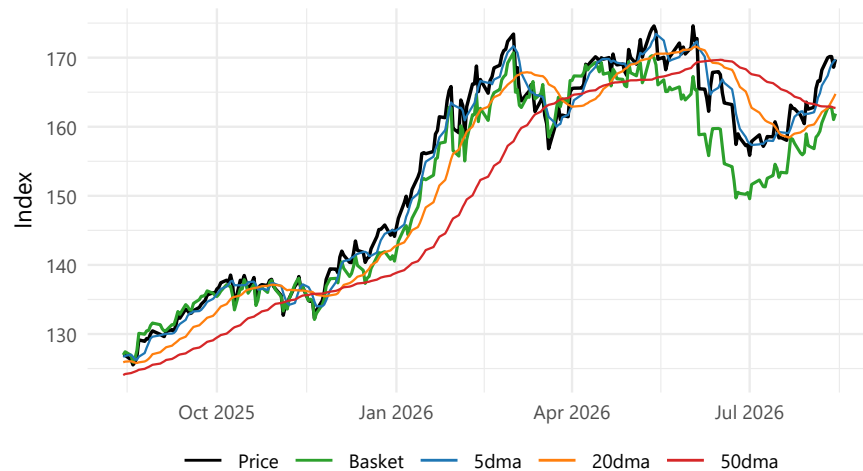
1	XLV	0.0623	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0605	iShares Biotechnology ETF	Healthcare	FLAT
6	FBT	0.0597	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0582	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	FLAT
7	XHE	0.0572	State Street SPDR S&P Health Care Equip...	Healthcare	FLAT
12	IHI	0.0565	iShares U.S. Medical Devices ETF	Healthcare	FLAT
17	PPH	0.0544	VanEck Pharmaceutical ETF	Healthcare	LONG
5	PTH	0.0536	Invesco Dorsey Wright Healthcare Moment...	Healthcare	LONG
4	XBI	0.0531	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0506	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0473	iShares U.S. Health Care Providers ETF	Healthcare	FLAT

Cluster 12 – Materials

Members: 17 **Var explained:** 0.684 **Signal:** LONG since 2026-08-05 (7 days, 1.9%)

Signal logic:

- Latest buy_cross: 2026-08-05 – filter **PASS** (252d at date = +34.1%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +32.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-05 > last sell at 2026-06-09)



Category breakdown

category	n_members	abs_weight
Materials	9	0.5595
MLP / Energy Infrastructure	4	0.2172
Energy	1	0.0645
Consumer Staples	1	0.0597
Norway Equity	1	0.0556
Industrial Metals	1	0.0435

Replication basket – AMLP+URA+GUNR (cor=0.98) basket lead at lag 1: 0.046

ticker	rep_wt	name	category	etf_signal
AMLP	0.138	Alerian MLP ETF	MLP / Energy Infrastr...	LONG
URA	0.095	Global X Uranium ETF	Materials	LONG
GUNR	0.775	FlexShares Global Upstream Natural Re...	Materials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	GNR	0.0693	State Street SPDR S&P Global Natural Re...	Materials	LONG
10	GUNR	0.0689	FlexShares Global Upstream Natural Reso...	Materials	LONG
17	NANR	0.0669	State Street SPDR S&P North American Na...	Materials	LONG
16	IGE	0.0645	iShares North American Natural Resource...	Energy	LONG
9	MXI	0.0634	iShares Global Materials ETF	Materials	LONG
12	PICK	0.0632	iShares MSCI Global Select Metals & Min...	Materials	LONG
3	XME	0.0613	State Street SPDR S&P Metals & Mining E...	Materials	LONG
13	SLX	0.0608	VanEck Steel ETF	Materials	LONG
5	MOO	0.0597	VanEck Agribusiness ETF	Consumer Staples	FLAT
2	COPX	0.0590	Global X Copper Miners ETF	Materials	LONG
8	MLPX	0.0564	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	FLAT
14	NORW	0.0556	Global X MSCI Norway ETF	Norway Equity	LONG
4	EMLP	0.0549	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
6	AMLP	0.0537	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
7	MLPA	0.0523	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
1	URA	0.0467	Global X Uranium ETF	Materials	LONG
15	DBB	0.0435	Invesco DB Base Metals Fund	Industrial Metals	LONG

Cluster 13 – China Equity

Members: 15 **Var explained:** 0.795 **Signal:** LONG since 2026-08-03 (9 days, -0.3%)

Signal logic:

- Latest buy_cross: 2026-08-03 – filter **PASS** (252d at date = +2.6%)
- Latest sell_cross: 2026-05-20
- Current 252d return: -2.3% – filter **BLOCKS** new entries today
- Decision: **LONG** (effective buy at 2026-08-03 > last sell at 2026-05-20)

**Category breakdown**

category	n_members	abs_weight
China Equity	13	0.8765
Emerging Markets Equity	1	0.0644
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
KWEB	0.562	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0740	WisdomTree China ex-State-Owned Enterpr...	China Equity	LONG
9	GXC	0.0738	State Street SPDR S&P China ETF	China Equity	LONG
1	MCHI	0.0732	iShares MSCI China ETF	China Equity	LONG
2	FXI	0.0712	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	FLAT
7	CHIQ	0.0705	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0680	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0668	Xtrackers Harvest CSI 300 China A-Share...	China Equity	FLAT
8	CNYA	0.0660	iShares MSCI China A ETF	China Equity	FLAT
13	KGRN	0.0653	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0644	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0597	KraneShares Boseri MSCI China A 50 Conn...	China Equity	LONG
12	EWI	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
14	CNXT	0.0588	VanEck ChiNext Innovators ETF	China Equity	FLAT
15	KURE	0.0582	KraneShares MSCI All China Health Care ...	China Equity	LONG

Cluster 14 – Financials

Members: 14 **Var explained:** 0.872 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +13.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.9382
Global Equity	1	0.0618

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.012

ticker	rep_wt	name	category	etf_signal
XLF	1.095	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0757	Vanguard Financials ETF	Financials	FLAT
11	FNCL	0.0756	Fidelity MSCI Financials Index ETF	Financials	LONG
7	IYF	0.0754	iShares U.S. Financial ETF	Financials	FLAT
5	FXO	0.0752	First Trust Financials AlphaDEX	Financials	LONG
8	IYG	0.0749	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0742	State Street Financial Select Sector SP...	Financials	LONG

9	KBWB	0.0733	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0713	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0707	State Street SPDR S&P Capital Markets E...	Financials	LONG
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	LONG
13	IAI	0.0696	iShares U.S. Broker-Dealers & Securitie...	Financials	FLAT
2	KRE	0.0688	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0630	State Street SPDR S&P Insurance ETF	Financials	FLAT
12	PSP	0.0618	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.961 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-08-06
- Current 252d return: +15.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-06 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: -0.001

ticker	rep_wt	name	category	etf_signal
VNQ	0.969	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

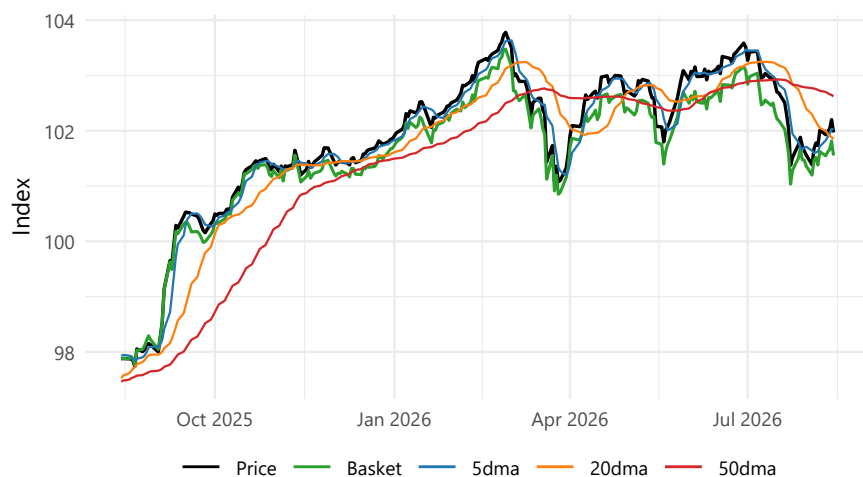
	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	FLAT
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	FLAT
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT
1	XLRE	0.0720	State Street Real Estate Select Sector ...	Real Estate	FLAT
5	ICF	0.0720	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0716	iShares Global REIT ETF	Real Estate	FLAT
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	FLAT
4	REZ	0.0696	iShares Residential and Multisector Rea...	Real Estate	FLAT
14	NURE	0.0687	Nuveen Short-Term REIT ETF	Real Estate	FLAT
12	INDS	0.0682	Pacer Industrial Real Estate ETF	Real Estate	FLAT

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.820 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-07-10
- Current 252d return: +4.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-10 more recent than last effective buy 2026-05-28)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.178

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	FLAT
6	CMF	0.0802	iShares California Muni Bond ETF	Municipal Bond	FLAT
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	FLAT
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	FLAT
1	FMB	0.0794	First Trust Managed Municipal ETF	Municipal Bond	FLAT
7	NYF	0.0794	iShares New York Muni Bond ETF	Municipal Bond	FLAT

5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	FLAT
4	MUNI	0.0770	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	FLAT
3	HYMB	0.0721	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	FLAT
12	SUB	0.0694	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 17 – Gold

Members: 12 **Var explained:** 0.863 **Signal:** LONG since 2026-08-07 (5 days, 0.9%)

Signal logic:

- Latest buy_cross: 2026-08-07 – filter **PASS** (252d at date = +36.2%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +35.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-07 > last sell at 2026-05-20)



Category breakdown

category	n_members	abs_weight
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Materials	5	0.4157
Gold	3	0.2502
Precious Metals	2	0.1723
Silver	2	0.1617

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.012

ticker	rep_wt	name	category	etf_signal
GLD	0.024	SPDR Gold Shares	Gold	LONG
IAU	0.750	iShares Gold Trust Shares	Gold	LONG
SLV	0.412	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0865	Invesco DB Precious Metals Fund	Precious Metals	LONG
7	GLTR	0.0858	abrdn Physical Precious Metals Basket S...	Precious Metals	LONG
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	LONG
2	GDXJ	0.0840	VanEck Junior Gold Miners ETF	Materials	LONG
3	GLD	0.0835	SPDR Gold Shares	Gold	LONG
9	IAU	0.0834	iShares Gold Trust Shares	Gold	LONG
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	LONG
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	LONG
6	SIL	0.0829	Global X Silver Miners ETF	Materials	LONG
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	LONG
4	SLV	0.0809	iShares Silver Trust	Silver	LONG
8	SIVR	0.0808	abrdn Physical Silver Shares ETF	Silver	LONG

Cluster 18 – Consumer Staples

Members: 11 **Var explained:** 0.742 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-26 – filter **PASS** (252d at date = +9.9%)
- Latest sell_cross: 2026-08-05
- Current 252d return: +4.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-05 more recent than last effective buy 2026-06-26)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5741
Utilities	4	0.3507
U.S. Small Cap Consumer Staples	1	0.0752

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.016

ticker	rep_wt	name	category	etf_signal
XLU	0.363	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.588	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0984	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0984	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0976	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0951	iShares U.S. Consumer Staples ETF	Consumer Staples	FLAT

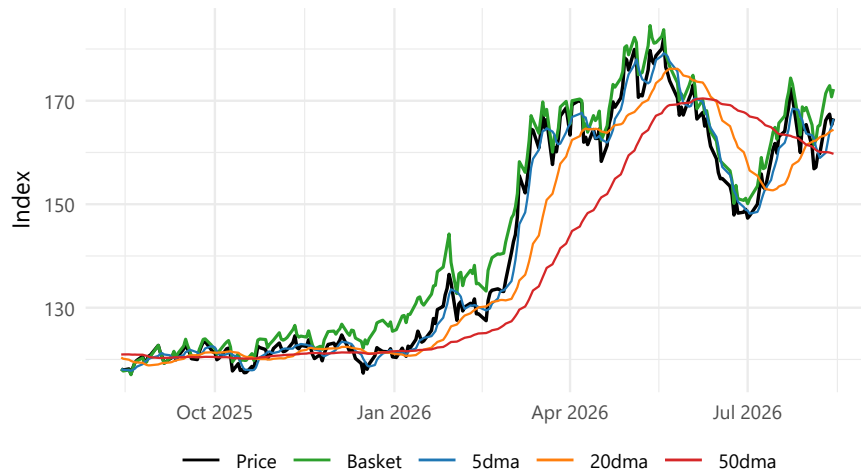
3	KXI	0.0947	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0900	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0887	iShares U.S. Utilities ETF	Utilities	FLAT
9	FUTY	0.0875	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
8	VPU	0.0874	Vanguard Utilities ETF	Utilities	FLAT
2	XLU	0.0871	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0752	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 19 – Broad Commodities

Members: 11 **Var explained:** 0.771 **Signal:** LONG since 2026-08-11 (3 days, -0.2%)

Signal logic:

- Latest buy_cross: 2026-08-11 – filter **PASS** (252d at date = +41.2%)
- Latest sell_cross: 2026-08-06
- Current 252d return: +41.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-11 > last sell at 2026-08-06)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6174
Oil	2	0.1980
Energy Commodities	1	0.1019
Agricultural Commodities	1	0.0494
Natural Gas	1	0.0333

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.078

ticker	rep_wt	name	category	etf_signal
PDBC	1.125	Invesco Optimum Yield Diversified Com...	Broad Commodities	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	DBC	0.1057	Invesco DB Commodity Index Tracking Fund	Broad Commodities	FLAT
6	GSG	0.1057	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
8	PDBC	0.1056	Invesco Optimum Yield Diversified Commo...	Broad Commodities	FLAT
9	COMT	0.1049	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
10	DBE	0.1019	Invesco DB Energy Fund	Energy Commodities	FLAT
1	USO	0.0996	United States Oil Fund	Oil	LONG
5	DBO	0.0984	Invesco DB Oil Fund	Oil	LONG
7	DJP	0.0984	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT
4	FTGC	0.0972	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT
3	DBA	0.0494	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
11	UNG	0.0333	United States Natural Gas Fund LP	Natural Gas	FLAT

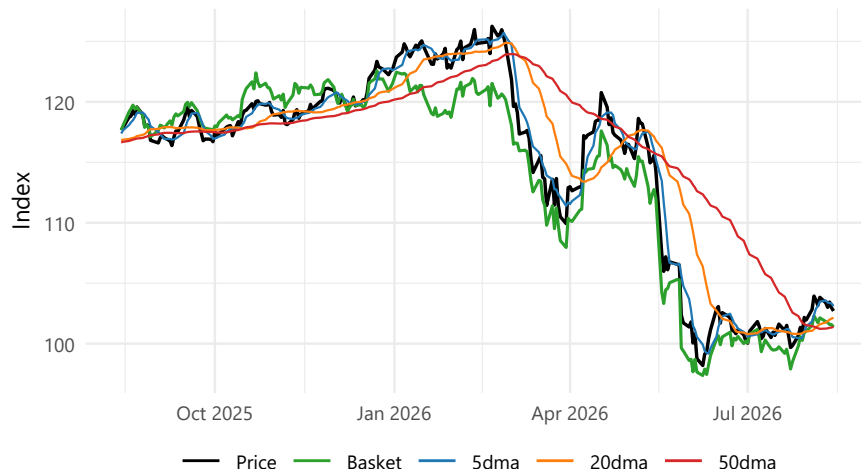
Cluster 20 – India Equity

Members: 10 **Var explained:** 0.405 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-08-03 – filter **BLOCKED** (252d at date = -10.3%; needs > 0)
- Latest effective buy: 2026-05-06 – delayed re-arm from blocked cross (252d at fire = +4.7%)
- Latest sell_cross: 2026-07-27
- Current 252d return: -13.3% – filter **BLOCKS** new entries today

- Decision: **FLAT** (last sell 2026-07-27 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4678
Malaysia Equity	1	0.1126
Indonesia Equity	1	0.1037
Philippines Equity	1	0.1014
Vietnam Equity	1	0.0751
Turkey Equity	1	0.0607
U.S. Dividend Equity	1	0.0547
Platinum	1	0.0239

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.005

ticker	rep_wt	name	category	etf_signal
NOBL	0.080	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.660	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.029	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1573	iShares MSCI India ETF	India Equity	FLAT
2	INDY	0.1556	iShares India 50 ETF	India Equity	FLAT
3	EPI	0.1549	WisdomTree India Earnings Fund	India Equity	FLAT
8	EWM	0.1126	iShares MSCI Malaysia Index Fund	Malaysia Equity	LONG
6	EIDO	0.1037	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1014	iShares MSCI Philippines ETF	Philippines Equity	LONG
10	VNM	0.0751	VanEck Vietnam ETF	Vietnam Equity	FLAT
9	TUR	0.0607	iShares MSCI Turkey ETF	Turkey Equity	LONG
4	NOBL	0.0547	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
5	PPLT	0.0239	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.904 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-07-02
- Current 252d return: -1.0% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-02 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
VTIP	0.268	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.760	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1036	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	FLAT
6	TIPX	0.1035	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1024	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDTT	0.1008	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.1000	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT

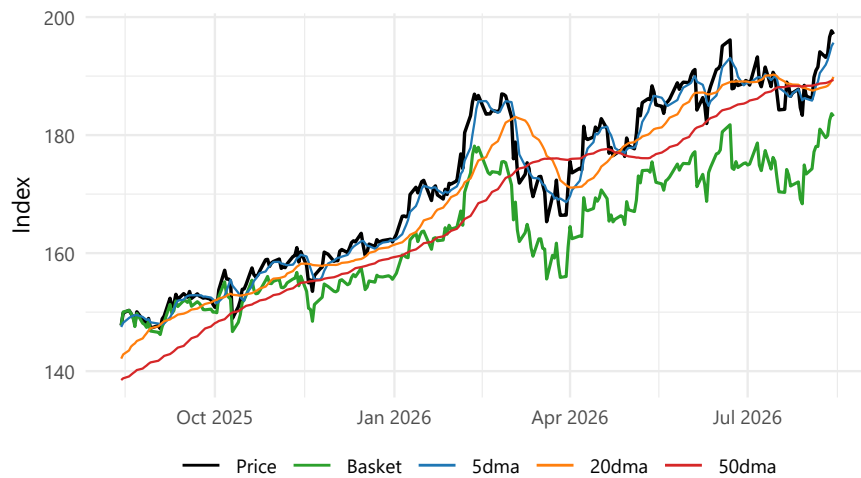
9	STIP	0.0998	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0989	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0884	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.887 **Signal:** LONG since 2026-08-06 (6 days, 2.9%)

Signal logic:

- Latest buy_cross: 2026-08-06 – filter **PASS** (252d at date = +35.6%)
- Latest sell_cross: 2026-07-28
- Current 252d return: +32.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-06 > last sell at 2026-07-28)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7805

Asia-Pacific Equity	2	0.2195
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Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.044

ticker	rep_wt	name	category	etf_signal
EWJ	0.907	iShares MSCI Japan Index Fund	Japan Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	LONG
8	BBJP	0.1142	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
6	DBJP	0.1118	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
9	HEWJ	0.1118	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
3	IPAC	0.1114	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	LONG
5	FLJH	0.1110	Franklin FTSE Japan Hedged ETF	Japan Equity	LONG
2	DXJ	0.1095	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
4	VPL	0.1081	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1079	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.555 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +3.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3622
Interest Rate Hedged Fixed Income	3	0.3243
Floating Rate Fixed Income	3	0.3135

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.293	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.217	Invesco Senior Loan ETF	Bank Loan / Floating ...	LONG
SRLN	0.298	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.123	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.035	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
2 SRLN	0.1251	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG

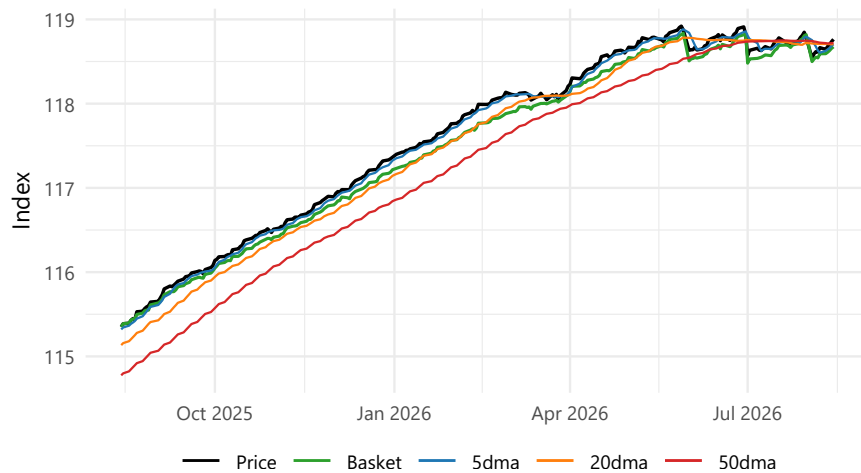
3	BKLN	0.1208	Invesco Senior Loan ETF	Bank Loan / Floating Rate	LONG
1	FTSL	0.1163	First Trust Senior Loan Fund	Bank Loan / Floating Rate	LONG
5	LQDH	0.1142	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1085	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1055	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1046	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1039	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1012	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 Var explained: 0.461 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-07-29 – filter **PASS** (252d at date = +3.3%)
- Latest sell_cross: 2026-08-05
- Current 252d return: +2.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-05 more recent than last effective buy 2026-07-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.86) basket lead at lag 1: 0.048

ticker	rep_wt	name	category	etf_signal
SHV	0.254	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.173	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	FLAT
GBIL	0.125	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	FLAT
ICSH	0.400	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1324	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1308	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1155	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	FLAT
5	GSY	0.1124	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	LONG
4	NEAR	0.1101	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
8	BIL	0.1065	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	FLAT
3	FTSM	0.1057	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
9	GBIL	0.0948	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	FLAT
1	VNLA	0.0917	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	FLAT

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.741 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-08-04 – filter **BLOCKED** (252d at date = -1.4%; needs > 0)
- Latest effective buy: 2026-05-29 – delayed re-arm from blocked cross (252d at fire = +0.6%)
- Latest sell_cross: 2026-07-23
- Current 252d return: -2.5% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-29)



Category breakdown

category	n_members	abs_weight
Currency	4	0.641
International Fixed Income	2	0.359

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.024

ticker	rep_wt	name	category	etf_signal
BWX	0.751	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1790	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1702	Invesco CurrencyShares Euro Currency Tr...	Currency	LONG
5	FXF	0.1651	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1609	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXY	0.1447	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse**Members:** 2 **Var explained:** 0.626 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-07-20 – filter **BLOCKED** (252d at date = -31.3%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-08-05
- Current 252d return: -34.1% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-08-05 more recent than last effective buy 2025-04-08)

**Category breakdown**

category	n_members	abs_weight
Currency	1	0.5
Leveraged / Inverse	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.022

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT
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Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
SIL	cluster_17	Gold	2026-08-07	5	1.2%	0.00	1.00
GDX	cluster_17	Gold	2026-08-06	6	7.2%	0.00	1.00
DBP	cluster_17	Gold	2026-08-07	5	0.9%	0.00	1.00
SLV	cluster_17	Gold	2026-08-10	4	-1.6%	0.00	1.00
SIVR	cluster_17	Gold	2026-08-10	4	-1.6%	0.00	1.00
GOAU	cluster_17	Gold	2026-08-06	6	6.5%	0.00	1.00
GLTR	cluster_17	Gold	2026-08-10	4	-0.9%	0.00	1.00
SGOL	cluster_17	Gold	2026-08-07	5	0.8%	0.00	1.00
GDXJ	cluster_17	Gold	2026-08-06	6	7.9%	0.00	1.00
IAU	cluster_17	Gold	2026-08-07	5	0.7%	0.00	1.00
GLD	cluster_17	Gold	2026-08-07	5	0.8%	0.00	1.00
SILJ	cluster_17	Gold	2026-08-07	5	2.5%	0.00	1.00
GNR	cluster_12	Materials	2026-07-27	14	5.1%	0.00	1.00
NANR	cluster_12	Materials	2026-07-24	15	6.4%	0.00	1.00
MXI	cluster_12	Materials	2026-08-05	7	-0.4%	0.00	1.00
IGE	cluster_12	Materials	2026-07-23	16	5.5%	0.00	1.00
PICK	cluster_12	Materials	2026-08-05	7	-1.9%	0.00	1.00
XME	cluster_12	Materials	2026-08-10	4	-0.3%	0.00	1.00
NORW	cluster_12	Materials	2026-08-05	7	5.0%	0.00	1.00
COPX	cluster_12	Materials	2026-08-05	7	-1.2%	0.00	1.00
SLX	cluster_12	Materials	2026-07-30	11	1.4%	0.00	1.00
URA	cluster_12	Materials	2026-08-11	3	-0.5%	0.00	1.00
GUNR	cluster_12	Materials	2026-07-27	14	3.1%	0.00	1.00
DBB	cluster_12	Materials	2026-08-04	8	0.5%	0.00	1.00
AML	cluster_12	Materials	2026-07-10	25	4.3%	0.88	0.12
MLPA	cluster_12	Materials	2026-07-10	25	4.4%	0.98	0.02
EMLP	cluster_12	Materials	2026-08-13	1	0.9%	0.74	-0.02
FNDX	cluster_4	U.S. Large Cap Value	2026-04-09	88	13.5%	0.00	1.00
FEX	cluster_4	U.S. Large Cap Value	2026-04-09	88	12.4%	0.00	1.00
VTV	cluster_4	U.S. Large Cap Value	2026-04-13	86	12.3%	0.00	1.00
DLN	cluster_4	U.S. Large Cap Value	2026-04-13	86	10.5%	0.00	1.00
IWD	cluster_4	U.S. Large Cap Value	2026-04-10	87	16.6%	0.00	1.00
DTD	cluster_4	U.S. Large Cap Value	2026-04-13	86	10.4%	0.00	1.00
VONV	cluster_4	U.S. Large Cap Value	2026-04-10	87	16.5%	0.00	1.00
PRF	cluster_4	U.S. Large Cap Value	2026-04-09	88	14.2%	0.00	1.00
RWL	cluster_4	U.S. Large Cap Value	2026-04-10	87	14.6%	0.00	1.00
PKW	cluster_4	U.S. Large Cap Value	2026-05-20	59	11.7%	0.00	1.00

JVAL	cluster_4	U.S. Large Cap Value	2026-07-30	11	4.9%	0.00	1.00
RSP	cluster_4	U.S. Large Cap Value	2026-04-14	85	11.6%	0.00	1.00
IYM	cluster_4	U.S. Large Cap Value	2026-08-06	6	2.0%	0.00	1.00
VIG	cluster_4	U.S. Large Cap Value	2026-04-14	85	9.6%	0.00	1.00
IWX	cluster_4	U.S. Large Cap Value	2026-04-10	87	17.4%	0.00	1.00
DGRO	cluster_4	U.S. Large Cap Value	2026-04-13	86	10.4%	0.00	1.00
IYZ	cluster_4	U.S. Large Cap Value	2026-08-06	6	0.9%	0.00	1.00
SCHV	cluster_4	U.S. Large Cap Value	2026-08-04	8	1.3%	0.00	1.00
VAW	cluster_4	U.S. Large Cap Value	2026-08-06	6	1.1%	0.00	1.00
AIVL	cluster_4	U.S. Large Cap Value	2026-04-10	87	13.1%	0.00	1.00
RSPM	cluster_4	U.S. Large Cap Value	2026-07-27	14	2.4%	0.00	1.00
XLB	cluster_4	U.S. Large Cap Value	2026-07-28	13	0.4%	0.00	1.00
FIDU	cluster_4	U.S. Large Cap Value	2026-08-05	7	0.4%	0.00	1.00
VLUE	cluster_4	U.S. Large Cap Value	2026-08-07	5	3.3%	0.00	1.00
VIS	cluster_4	U.S. Large Cap Value	2026-08-05	7	0.3%	0.00	1.00
PAVE	cluster_4	U.S. Large Cap Value	2026-08-05	7	-0.1%	0.00	0.99
ITB	cluster_4	U.S. Large Cap Value	2026-07-28	13	-1.5%	0.04	0.63
XHB	cluster_4	U.S. Large Cap Value	2026-08-07	5	-1.6%	0.05	0.38
IQLT	cluster_1	Developed ex-US Equity	2026-07-01	31	5.8%	0.00	1.00
FNDF	cluster_1	Developed ex-US Equity	2026-07-31	10	3.1%	0.00	1.00
TLTD	cluster_1	Developed ex-US Equity	2026-07-30	11	3.0%	0.00	1.00
SCHC	cluster_1	Developed ex-US Equity	2026-08-05	7	2.3%	0.00	1.00
PXF	cluster_1	Developed ex-US Equity	2026-08-03	9	3.1%	0.00	1.00
FNDC	cluster_1	Developed ex-US Equity	2026-08-04	8	2.0%	0.00	1.00
EWC	cluster_1	Developed ex-US Equity	2026-07-10	25	6.1%	0.00	1.00
INTF	cluster_1	Developed ex-US Equity	2026-07-27	14	4.5%	0.00	1.00
HEZU	cluster_1	Developed ex-US Equity	2026-07-30	11	3.3%	0.00	1.00
RODM	cluster_1	Developed ex-US Equity	2026-07-27	14	2.6%	0.00	1.00
GWX	cluster_1	Developed ex-US Equity	2026-08-06	6	2.9%	0.00	1.00
DLS	cluster_1	Developed ex-US Equity	2026-08-03	9	2.8%	0.00	1.00
SPDW	cluster_1	Developed ex-US Equity	2026-08-05	7	1.8%	0.00	1.00
VEA	cluster_1	Developed ex-US Equity	2026-08-05	7	1.7%	0.00	1.00
GSIE	cluster_1	Developed ex-US Equity	2026-07-02	30	5.1%	0.00	1.00
SCHF	cluster_1	Developed ex-US Equity	2026-08-05	7	1.6%	0.00	1.00
ESGD	cluster_1	Developed ex-US Equity	2026-07-30	11	2.3%	0.00	1.00
EFA	cluster_1	Developed ex-US Equity	2026-07-30	11	2.3%	0.00	1.00
SCZ	cluster_1	Developed ex-US Equity	2026-08-04	8	2.0%	0.00	1.00
EZU	cluster_1	Developed ex-US Equity	2026-07-31	10	3.5%	0.00	1.00
FEZ	cluster_1	Developed ex-US Equity	2026-07-30	11	3.4%	0.00	1.00
IEFA	cluster_1	Developed ex-US Equity	2026-07-30	11	2.3%	0.00	1.00
EPOL	cluster_1	Developed ex-US Equity	2026-07-08	27	11.7%	0.00	1.00
EWS	cluster_1	Developed ex-US Equity	2026-06-17	40	13.8%	0.00	1.00
EFG	cluster_1	Developed ex-US Equity	2026-08-05	7	1.6%	0.00	1.00
QEFA	cluster_1	Developed ex-US Equity	2026-07-06	29	3.5%	0.00	1.00
IXUS	cluster_1	Developed ex-US Equity	2026-08-05	7	1.4%	0.00	1.00

VXUS	cluster_1	Developed	ex-US	Equity	2026-08-05	7	1.4%	0.00	1.00
CWI	cluster_1	Developed	ex-US	Equity	2026-08-04	8	1.2%	0.00	1.00
ACWX	cluster_1	Developed	ex-US	Equity	2026-08-05	7	1.2%	0.00	1.00
VEU	cluster_1	Developed	ex-US	Equity	2026-08-05	7	1.3%	0.00	1.00
EZA	cluster_1	Developed	ex-US	Equity	2026-08-05	7	1.0%	0.00	1.00
DWM	cluster_1	Developed	ex-US	Equity	2026-07-30	11	1.8%	0.00	1.00
VIGI	cluster_1	Developed	ex-US	Equity	2026-06-30	32	6.5%	0.00	1.00
FEP	cluster_1	Developed	ex-US	Equity	2026-07-31	10	1.9%	0.00	1.00
VSS	cluster_1	Developed	ex-US	Equity	2026-08-07	5	0.8%	0.00	1.00
JPIN	cluster_1	Developed	ex-US	Equity	2026-07-27	14	3.3%	0.00	1.00
EWG	cluster_1	Developed	ex-US	Equity	2026-07-31	10	3.1%	0.00	1.00
IPKW	cluster_1	Developed	ex-US	Equity	2026-07-10	25	5.5%	0.00	1.00
VGK	cluster_1	Developed	ex-US	Equity	2026-07-01	31	5.2%	0.00	1.00
HEDJ	cluster_1	Developed	ex-US	Equity	2026-08-03	9	1.1%	0.00	1.00
IEUR	cluster_1	Developed	ex-US	Equity	2026-07-30	11	1.5%	0.00	1.00
IEV	cluster_1	Developed	ex-US	Equity	2026-07-27	14	3.8%	0.00	1.00
HFXI	cluster_1	Developed	ex-US	Equity	2026-08-05	7	1.7%	0.00	1.00
IVLU	cluster_1	Developed	ex-US	Equity	2026-07-27	14	3.6%	0.00	1.00
EWQ	cluster_1	Developed	ex-US	Equity	2026-07-30	11	2.2%	0.00	1.00
IQDF	cluster_1	Developed	ex-US	Equity	2026-08-04	8	0.5%	0.00	1.00
VYMI	cluster_1	Developed	ex-US	Equity	2026-07-08	27	5.1%	0.00	1.00
IMTM	cluster_1	Developed	ex-US	Equity	2026-08-05	7	2.1%	0.00	1.00
EFV	cluster_1	Developed	ex-US	Equity	2026-07-08	27	5.6%	0.00	1.00
VIDI	cluster_1	Developed	ex-US	Equity	2026-08-10	4	0.9%	0.00	1.00
EUFN	cluster_1	Developed	ex-US	Equity	2026-06-12	43	9.6%	0.00	1.00
EBND	cluster_1	Developed	ex-US	Equity	2026-07-31	10	0.7%	0.00	1.00
EWN	cluster_1	Developed	ex-US	Equity	2026-08-05	7	4.1%	0.00	1.00
EWO	cluster_1	Developed	ex-US	Equity	2026-07-31	10	4.3%	0.00	1.00
EMLC	cluster_1	Developed	ex-US	Equity	2026-08-05	7	0.2%	0.00	1.00
EPP	cluster_1	Developed	ex-US	Equity	2026-07-15	22	2.9%	0.01	0.99
FXA	cluster_1	Developed	ex-US	Equity	2026-08-04	8	0.6%	0.01	0.99
EWD	cluster_1	Developed	ex-US	Equity	2026-07-30	11	0.8%	0.01	0.99
IDV	cluster_1	Developed	ex-US	Equity	2026-07-28	13	2.7%	0.01	0.99
EFAV	cluster_1	Developed	ex-US	Equity	2026-07-16	21	3.8%	0.02	0.98
EWA	cluster_1	Developed	ex-US	Equity	2026-07-20	19	4.0%	0.02	0.98
EWU	cluster_1	Developed	ex-US	Equity	2026-07-16	21	2.7%	0.04	0.96
EWL	cluster_1	Developed	ex-US	Equity	2026-07-28	13	0.1%	0.05	0.95
BBAX	cluster_1	Developed	ex-US	Equity	2026-07-14	23	3.3%	0.05	0.95
ECH	cluster_1	Developed	ex-US	Equity	2026-08-05	7	-1.6%	0.05	0.92
NFRA	cluster_1	Developed	ex-US	Equity	2026-07-16	21	1.4%	0.23	0.77
HDEF	cluster_1	Developed	ex-US	Equity	2026-07-09	26	3.7%	0.28	0.72
DWX	cluster_1	Developed	ex-US	Equity	2026-07-09	26	2.8%	0.37	0.63
IDLV	cluster_1	Developed	ex-US	Equity	2026-07-10	25	2.9%	0.48	0.52
ENZL	cluster_1	Developed	ex-US	Equity	2026-07-07	28	4.2%	0.55	0.45
EWK	cluster_1	Developed	ex-US	Equity	2026-07-29	12	-1.6%	0.43	0.25

RWX	cluster_1	Developed ex-US Equity	2026-07-16	21	0.9%	0.76	0.24
EWX	cluster_1	Developed ex-US Equity	2026-07-31	10	-2.3%	0.71	0.18
IYH	cluster_11	Healthcare	2026-05-20	59	14.2%	0.00	1.00
VHT	cluster_11	Healthcare	2026-05-19	60	14.8%	0.00	1.00
FHLC	cluster_11	Healthcare	2026-05-19	60	14.8%	0.00	1.00
XLV	cluster_11	Healthcare	2026-05-21	58	13.0%	0.00	1.00
PTH	cluster_11	Healthcare	2026-06-17	40	14.3%	0.03	0.97
PPH	cluster_11	Healthcare	2026-05-21	58	6.8%	0.12	0.88
IPAC	cluster_22	Japan Equity	2026-08-03	9	4.1%	0.00	1.00
BBJP	cluster_22	Japan Equity	2026-08-05	7	3.1%	0.00	1.00
EWJ	cluster_22	Japan Equity	2026-08-05	7	3.2%	0.00	1.00
EWJV	cluster_22	Japan Equity	2026-07-06	29	4.1%	0.00	1.00
DXJ	cluster_22	Japan Equity	2026-08-05	7	3.4%	0.00	1.00
DBJP	cluster_22	Japan Equity	2026-08-10	4	2.2%	0.00	1.00
HEWJ	cluster_22	Japan Equity	2026-08-07	5	2.5%	0.00	1.00
FLJH	cluster_22	Japan Equity	2026-08-10	4	2.2%	0.00	1.00
VPL	cluster_22	Japan Equity	2026-08-10	4	3.4%	0.00	0.99
SMLF	cluster_2	U.S. Mid Cap Equity	2026-08-03	9	3.3%	0.00	1.00
ISCG	cluster_2	U.S. Mid Cap Equity	2026-08-04	8	1.7%	0.00	1.00
IVOG	cluster_2	U.S. Mid Cap Equity	2026-08-06	6	3.3%	0.00	1.00
MDYG	cluster_2	U.S. Mid Cap Equity	2026-08-06	6	3.4%	0.00	1.00
VXF	cluster_2	U.S. Mid Cap Equity	2026-08-05	7	2.8%	0.00	1.00
IJK	cluster_2	U.S. Mid Cap Equity	2026-08-06	6	3.4%	0.00	1.00
VTWO	cluster_2	U.S. Mid Cap Equity	2026-08-04	8	1.1%	0.00	1.00
IWM	cluster_2	U.S. Mid Cap Equity	2026-08-04	8	1.1%	0.00	1.00
VBK	cluster_2	U.S. Mid Cap Equity	2026-08-07	5	1.5%	0.00	1.00
IWC	cluster_2	U.S. Mid Cap Equity	2026-08-05	7	2.2%	0.00	1.00
SDVY	cluster_2	U.S. Mid Cap Equity	2026-06-08	47	8.3%	0.00	1.00
IJS	cluster_2	U.S. Mid Cap Equity	2026-04-10	87	14.5%	0.00	1.00
ISMD	cluster_2	U.S. Mid Cap Equity	2026-04-09	88	19.9%	0.00	1.00
IWO	cluster_2	U.S. Mid Cap Equity	2026-08-06	6	3.0%	0.00	1.00
VTWG	cluster_2	U.S. Mid Cap Equity	2026-08-06	6	2.9%	0.00	1.00
RSPD	cluster_2	U.S. Mid Cap Equity	2026-07-28	13	0.7%	0.00	1.00
SCHM	cluster_2	U.S. Mid Cap Equity	2026-08-07	5	1.7%	0.00	1.00
IYC	cluster_2	U.S. Mid Cap Equity	2026-08-04	8	0.6%	0.00	0.99
RTH	cluster_2	U.S. Mid Cap Equity	2026-07-29	12	2.4%	0.01	0.99
REGL	cluster_2	U.S. Mid Cap Equity	2026-06-12	43	5.8%	0.03	0.97
SCHA	cluster_2	U.S. Mid Cap Equity	2026-08-07	5	2.4%	0.00	0.95
XLV	cluster_2	U.S. Mid Cap Equity	2026-08-05	7	-0.4%	0.02	0.95
FDIS	cluster_2	U.S. Mid Cap Equity	2026-08-05	7	-0.7%	0.03	0.92
VCR	cluster_2	U.S. Mid Cap Equity	2026-08-05	7	-0.7%	0.04	0.91
SMMV	cluster_2	U.S. Mid Cap Equity	2026-06-25	35	6.0%	0.10	0.90
XSMO	cluster_2	U.S. Mid Cap Equity	2026-08-13	1	0.9%	0.00	0.65
REM	cluster_2	U.S. Mid Cap Equity	2026-08-12	2	0.6%	0.19	0.57
XRT	cluster_2	U.S. Mid Cap Equity	2026-05-28	54	3.9%	0.79	0.02

IYT	cluster_2	U.S. Mid Cap Equity	2026-08-06	6	1.4%	0.54	-0.16
FTXR	cluster_2	U.S. Mid Cap Equity	2026-08-13	1	0.1%	0.55	-0.37
SKYY	cluster_10	Technology	2026-07-30	11	18.1%	0.00	1.00
FDN	cluster_10	Technology	2026-08-03	9	3.0%	0.00	1.00
ARKQ	cluster_10	Technology	2026-08-11	3	1.6%	0.00	1.00
ARKK	cluster_10	Technology	2026-08-10	4	0.8%	0.00	0.99
XLC	cluster_10	Technology	2026-08-06	6	1.6%	0.11	0.86
IPO	cluster_10	Technology	2026-08-13	1	-0.3%	0.00	0.82
FCOM	cluster_10	Technology	2026-08-13	1	0.3%	0.26	0.42
VOX	cluster_10	Technology	2026-08-13	1	0.4%	0.27	0.40
XAR	cluster_3	U.S. Large Cap Equity	2026-08-06	6	4.9%	0.00	1.00
PRNT	cluster_3	U.S. Large Cap Equity	2026-08-05	7	2.7%	0.00	1.00
IMCG	cluster_3	U.S. Large Cap Equity	2026-08-04	8	1.9%	0.00	1.00
QQEW	cluster_3	U.S. Large Cap Equity	2026-08-04	8	2.5%	0.00	1.00
JHML	cluster_3	U.S. Large Cap Equity	2026-08-03	9	2.5%	0.00	1.00
PPA	cluster_3	U.S. Large Cap Equity	2026-07-27	14	4.5%	0.00	1.00
VOT	cluster_3	U.S. Large Cap Equity	2026-08-05	7	2.8%	0.00	1.00
SPGM	cluster_3	U.S. Large Cap Equity	2026-08-04	8	1.1%	0.00	1.00
DGRW	cluster_3	U.S. Large Cap Equity	2026-08-03	9	2.3%	0.00	1.00
VT	cluster_3	U.S. Large Cap Equity	2026-08-04	8	1.1%	0.00	1.00
AOA	cluster_3	U.S. Large Cap Equity	2026-08-05	7	0.7%	0.00	1.00
ACWI	cluster_3	U.S. Large Cap Equity	2026-08-04	8	1.0%	0.00	1.00
GSLC	cluster_3	U.S. Large Cap Equity	2026-08-03	9	2.7%	0.00	1.00
AOR	cluster_3	U.S. Large Cap Equity	2026-08-05	7	0.6%	0.00	1.00
URTH	cluster_3	U.S. Large Cap Equity	2026-08-03	9	2.6%	0.00	1.00
LRGF	cluster_3	U.S. Large Cap Equity	2026-08-03	9	2.9%	0.00	1.00
IWP	cluster_3	U.S. Large Cap Equity	2026-08-07	5	2.2%	0.00	1.00
ITA	cluster_3	U.S. Large Cap Equity	2026-07-24	15	5.4%	0.00	1.00
EPS	cluster_3	U.S. Large Cap Equity	2026-08-03	9	2.4%	0.00	1.00
CRBN	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.7%	0.00	1.00
VTI	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.8%	0.00	1.00
SUSA	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.9%	0.00	1.00
SCHB	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.8%	0.00	1.00
IWV	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.8%	0.00	1.00
ITOT	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.8%	0.00	1.00
SCHX	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.7%	0.00	1.00
VTHR	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.7%	0.00	1.00
VONE	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.8%	0.00	1.00
IWB	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.8%	0.00	1.00
IYY	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.7%	0.00	1.00
QAI	cluster_3	U.S. Large Cap Equity	2026-08-06	6	1.1%	0.00	1.00
SPTM	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.7%	0.00	1.00
ILCB	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.8%	0.00	1.00
VV	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.7%	0.00	1.00
VOO	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.7%	0.00	1.00

CATH	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.6%	0.00	1.00
IVV	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.7%	0.00	1.00
SPY	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.6%	0.00	1.00
DSI	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.9%	0.00	1.00
AOM	cluster_3	U.S. Large Cap Equity	2026-08-06	6	0.6%	0.00	1.00
BOTZ	cluster_3	U.S. Large Cap Equity	2026-08-07	5	0.3%	0.00	1.00
ESGU	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.6%	0.00	1.00
EIS	cluster_3	U.S. Large Cap Equity	2026-08-12	2	1.3%	0.00	1.00
SPYX	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.5%	0.00	1.00
IWL	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.6%	0.00	1.00
MGC	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.5%	0.00	1.00
IOO	cluster_3	U.S. Large Cap Equity	2026-08-03	9	1.4%	0.00	1.00
ROBO	cluster_3	U.S. Large Cap Equity	2026-08-07	5	0.5%	0.00	1.00
XT	cluster_3	U.S. Large Cap Equity	2026-08-07	5	0.5%	0.00	1.00
OEF	cluster_3	U.S. Large Cap Equity	2026-08-04	8	0.2%	0.00	1.00
AOK	cluster_3	U.S. Large Cap Equity	2026-08-06	6	0.4%	0.00	0.99
FV	cluster_3	U.S. Large Cap Equity	2026-08-10	4	1.2%	0.00	0.99
XLG	cluster_3	U.S. Large Cap Equity	2026-08-05	7	0.3%	0.01	0.99
QMOM	cluster_3	U.S. Large Cap Equity	2026-08-12	2	0.4%	0.00	0.96
FTC	cluster_3	U.S. Large Cap Equity	2026-08-10	4	2.1%	0.00	0.95
SPHB	cluster_3	U.S. Large Cap Equity	2026-08-12	2	1.1%	0.00	0.91
CWB	cluster_3	U.S. Large Cap Equity	2026-08-14	0	0.0%	0.00	0.84
VFMO	cluster_3	U.S. Large Cap Equity	2026-08-13	1	0.9%	0.00	0.74
SPHQ	cluster_3	U.S. Large Cap Equity	2026-08-14	0	0.0%	0.00	0.55
DWAS	cluster_3	U.S. Large Cap Equity	2026-08-14	0	0.0%	0.02	0.49
KCE	cluster_14	Financials	2026-07-07	28	7.4%	0.00	1.00
XLF	cluster_14	Financials	2026-04-13	86	12.6%	0.00	1.00
FNCL	cluster_14	Financials	2026-04-13	86	12.8%	0.01	0.99
IAT	cluster_14	Financials	2026-05-20	59	15.4%	0.04	0.96
FXO	cluster_14	Financials	2026-06-05	48	12.0%	0.22	0.78
TUR	cluster_20	India Equity	2026-08-14	0	0.0%	0.14	0.61
EPHE	cluster_20	India Equity	2026-08-03	9	-1.5%	0.44	0.56
EWM	cluster_20	India Equity	2026-07-30	11	-0.9%	0.77	0.23
SCHD	cluster_8	U.S. Dividend Equity	2026-07-07	28	6.1%	0.00	1.00
ACWV	cluster_8	U.S. Dividend Equity	2026-07-06	29	3.8%	0.00	1.00
FTA	cluster_8	U.S. Dividend Equity	2026-05-21	58	10.7%	0.00	1.00
PWV	cluster_8	U.S. Dividend Equity	2026-04-02	92	17.0%	0.00	1.00
FTCS	cluster_8	U.S. Dividend Equity	2026-06-26	34	7.5%	0.00	1.00
HDV	cluster_8	U.S. Dividend Equity	2026-06-25	35	6.6%	0.01	0.99
OUSA	cluster_8	U.S. Dividend Equity	2026-07-01	31	5.2%	0.01	0.99
SDY	cluster_8	U.S. Dividend Equity	2026-05-20	59	7.0%	0.01	0.99
FDL	cluster_8	U.S. Dividend Equity	2026-07-16	21	3.2%	0.02	0.98
FVD	cluster_8	U.S. Dividend Equity	2026-06-10	45	6.6%	0.07	0.93
LGLV	cluster_8	U.S. Dividend Equity	2026-06-26	34	3.6%	0.28	0.72
SPHD	cluster_8	U.S. Dividend Equity	2026-06-08	47	6.7%	0.64	0.36

KURE	cluster_13	China Equity	2026-07-06	29	7.5%	0.00	1.00
CXSE	cluster_13	China Equity	2026-08-07	5	-2.5%	0.03	0.94
GXC	cluster_13	China Equity	2026-07-31	10	-1.2%	0.10	0.90
MCHI	cluster_13	China Equity	2026-08-04	8	-2.6%	0.42	0.58
KBA	cluster_13	China Equity	2026-08-12	2	-1.1%	0.31	0.23
FXE	cluster_25	International Fixed Inc...	2026-08-04	8	0.4%	0.01	0.99
FXB	cluster_25	International Fixed Inc...	2026-07-31	10	0.3%	0.09	0.91
SCHE	cluster_7	Emerging Markets Equity	2026-08-05	7	0.1%	0.00	1.00
VWO	cluster_7	Emerging Markets Equity	2026-08-06	6	0.3%	0.00	1.00
GMF	cluster_7	Emerging Markets Equity	2026-08-06	6	1.1%	0.00	1.00
SPEM	cluster_7	Emerging Markets Equity	2026-08-06	6	0.5%	0.00	1.00
DEM	cluster_7	Emerging Markets Equity	2026-08-07	5	-0.1%	0.00	0.99
EEMV	cluster_7	Emerging Markets Equity	2026-08-12	2	0.4%	0.00	0.99
PXH	cluster_7	Emerging Markets Equity	2026-07-31	10	-0.1%	0.01	0.99
FEM	cluster_7	Emerging Markets Equity	2026-08-07	5	0.6%	0.00	0.98
FNDE	cluster_7	Emerging Markets Equity	2026-07-30	11	0.7%	0.03	0.97
EEMA	cluster_7	Emerging Markets Equity	2026-08-12	2	0.2%	0.00	0.97
EWT	cluster_7	Emerging Markets Equity	2026-08-11	3	3.0%	0.00	0.93
IEMG	cluster_7	Emerging Markets Equity	2026-08-13	1	-0.1%	0.00	0.88
GEM	cluster_7	Emerging Markets Equity	2026-08-13	1	-0.0%	0.00	0.87
FEMS	cluster_7	Emerging Markets Equity	2026-08-12	2	-0.4%	0.01	0.87
ESGE	cluster_7	Emerging Markets Equity	2026-08-14	0	0.0%	0.00	0.84
DGS	cluster_7	Emerging Markets Equity	2026-08-12	2	-0.9%	0.00	0.83
EMGF	cluster_7	Emerging Markets Equity	2026-08-14	0	0.0%	0.00	0.82
AIA	cluster_7	Emerging Markets Equity	2026-08-14	0	0.0%	0.00	0.79
EEM	cluster_7	Emerging Markets Equity	2026-08-14	0	0.0%	0.00	0.79
AAXJ	cluster_7	Emerging Markets Equity	2026-08-14	0	0.0%	0.00	0.78
EWX	cluster_7	Emerging Markets Equity	2026-08-14	0	0.0%	0.00	0.75
DVYE	cluster_7	Emerging Markets Equity	2026-07-24	15	0.3%	0.39	0.61
SDIV	cluster_7	Emerging Markets Equity	2026-08-13	1	-0.2%	1.00	-0.52
FXL	cluster_6	U.S. Large Cap Growth	2026-08-06	6	5.8%	0.00	1.00
TDIV	cluster_6	U.S. Large Cap Growth	2026-08-07	5	0.0%	0.00	1.00
PTLC	cluster_6	U.S. Large Cap Growth	2026-08-04	8	0.6%	0.00	1.00
VGT	cluster_6	U.S. Large Cap Growth	2026-08-06	6	2.5%	0.00	1.00
FTEC	cluster_6	U.S. Large Cap Growth	2026-08-06	6	2.5%	0.00	1.00
SCHG	cluster_6	U.S. Large Cap Growth	2026-08-04	8	0.8%	0.00	1.00
NULG	cluster_6	U.S. Large Cap Growth	2026-08-05	7	0.9%	0.00	1.00
IUSG	cluster_6	U.S. Large Cap Growth	2026-08-05	7	0.7%	0.00	1.00
SPYG	cluster_6	U.S. Large Cap Growth	2026-08-05	7	0.5%	0.00	1.00
IVW	cluster_6	U.S. Large Cap Growth	2026-08-05	7	0.5%	0.00	1.00
VOOG	cluster_6	U.S. Large Cap Growth	2026-08-05	7	0.6%	0.00	1.00
NXTG	cluster_6	U.S. Large Cap Growth	2026-08-12	2	0.7%	0.00	0.99
VUG	cluster_6	U.S. Large Cap Growth	2026-08-05	7	0.7%	0.01	0.99
ONEQ	cluster_6	U.S. Large Cap Growth	2026-08-06	6	1.4%	0.00	0.99
IGM	cluster_6	U.S. Large Cap Growth	2026-08-06	6	1.8%	0.00	0.99

ILCG	cluster_6	U.S. Large Cap Growth	2026-08-06	6	1.4%	0.00	0.99
MGK	cluster_6	U.S. Large Cap Growth	2026-08-05	7	0.7%	0.01	0.99
IYW	cluster_6	U.S. Large Cap Growth	2026-08-06	6	2.0%	0.00	0.99
XLK	cluster_6	U.S. Large Cap Growth	2026-08-07	5	1.1%	0.00	0.99
VONG	cluster_6	U.S. Large Cap Growth	2026-08-06	6	1.4%	0.01	0.98
IWF	cluster_6	U.S. Large Cap Growth	2026-08-06	6	1.4%	0.01	0.98
AIQ	cluster_6	U.S. Large Cap Growth	2026-08-10	4	1.1%	0.00	0.98
IXN	cluster_6	U.S. Large Cap Growth	2026-08-07	5	1.7%	0.00	0.97
IWY	cluster_6	U.S. Large Cap Growth	2026-08-07	5	0.5%	0.01	0.96
QQQ	cluster_6	U.S. Large Cap Growth	2026-08-07	5	1.1%	0.00	0.94
QTUM	cluster_6	U.S. Large Cap Growth	2026-08-13	1	0.3%	0.00	0.94
QTEC	cluster_6	U.S. Large Cap Growth	2026-08-10	4	3.4%	0.00	0.93
PTNQ	cluster_6	U.S. Large Cap Growth	2026-08-07	5	1.0%	0.00	0.93
QYLD	cluster_6	U.S. Large Cap Growth	2026-08-07	5	0.4%	0.01	0.85
XNTK	cluster_6	U.S. Large Cap Growth	2026-08-13	1	-0.7%	0.00	0.79
PWB	cluster_6	U.S. Large Cap Growth	2026-08-07	5	1.7%	0.01	0.76
PTF	cluster_6	U.S. Large Cap Growth	2026-04-07	90	22.2%	0.00	0.23
SRLN	cluster_23	Bank Loan / Floating Ra...	2026-07-30	11	0.4%	0.00	1.00
BKLN	cluster_23	Bank Loan / Floating Ra...	2026-08-05	7	0.4%	0.00	1.00
FTSL	cluster_23	Bank Loan / Floating Ra...	2026-08-07	5	0.1%	0.02	0.94
USO	cluster_19	Broad Commodities	2026-08-11	3	-0.8%	0.63	0.36
DBO	cluster_19	Broad Commodities	2026-08-11	3	-0.6%	0.69	0.29
PFXF	cluster_9	High Yield Bond	2026-08-10	4	1.3%	0.00	1.00
PFF	cluster_9	High Yield Bond	2026-08-13	1	-0.2%	0.00	0.99
HYLS	cluster_9	High Yield Bond	2026-08-13	1	-0.1%	0.05	0.60
GSY	cluster_24	Cash / Ultra-Short Fixe...	2026-08-07	5	0.1%	0.03	0.82
SUB	cluster_16	Municipal Bond	2026-08-13	1	-0.0%	0.00	0.71
BSCQ	cluster_5	Aggregate Fixed Income	2026-08-07	5	0.1%	0.03	0.86
LDUR	cluster_5	Aggregate Fixed Income	2026-07-31	10	-0.0%	0.25	0.34
FSTA	cluster_18	Consumer Staples	2026-07-28	13	-1.0%	0.28	0.68
VDC	cluster_18	Consumer Staples	2026-07-28	13	-0.9%	0.29	0.66
XLP	cluster_18	Consumer Staples	2026-07-27	14	0.9%	0.42	0.50
PSCC	cluster_18	Consumer Staples	2026-06-10	45	9.4%	0.54	0.46
KXI	cluster_18	Consumer Staples	2026-07-28	13	-1.2%	0.52	0.39

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
VO	cluster_4	U.S. Large Cap Value	1.00	1.00
HEFA	cluster_1	Developed ex-US Equity	1.00	1.00
JHMM	cluster_2	U.S. Mid Cap Equity	1.00	1.00
GNR	cluster_12	Materials	1.00	1.00
SIL	cluster_17	Gold	1.00	1.00
SKYY	cluster_10	Technology	1.00	1.00
PSP	cluster_14	Financials	1.00	1.00
XAR	cluster_3	U.S. Large Cap Equity	1.00	1.00
IHI	cluster_11	Healthcare	1.00	1.00
SCHD	cluster_8	U.S. Dividend Equity	1.00	1.00
IPAC	cluster_22	Japan Equity	1.00	1.00
FXL	cluster_6	U.S. Large Cap Growth	1.00	1.00
PFXF	cluster_9	High Yield Bond	1.00	1.00
SRLN	cluster_23	Bank Loan / Floating Rate	1.00	1.00
SCHE	cluster_7	Emerging Markets Equity	1.00	1.00
COMT	cluster_19	Broad Commodities	1.00	0.95
BSCQ	cluster_5	Aggregate Fixed Income	0.89	0.86
GSY	cluster_24	Cash / Ultra-Short Fixed ...	0.86	0.82
SUB	cluster_16	Municipal Bond	0.71	0.71
FSTA	cluster_18	Consumer Staples	0.96	0.68

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
PSK	cluster_9	High Yield Bond	1.00	-1.00
NURE	cluster_15	Real Estate	1.00	-1.00
IGF	cluster_1	Developed ex-US Equity	1.00	-1.00
SCHP	cluster_21	Inflation-Protected Fixed...	0.99	-0.99
XLU	cluster_18	Consumer Staples	0.99	-0.99
TLT	cluster_5	Aggregate Fixed Income	0.98	-0.98

SQQQ	cluster_26	Leveraged / Inverse	1.00	-0.96
XTN	cluster_2	U.S. Mid Cap Equity	0.94	-0.94
FLOT	cluster_23	Bank Loan / Floating Rate	0.95	-0.91
MLPX	cluster_12	Materials	0.96	-0.82
LVHD	cluster_8	U.S. Dividend Equity	0.99	-0.75
BIL	cluster_24	Cash / Ultra-Short Fixed ...	0.81	-0.52
SDIV	cluster_7	Emerging Markets Equity	1.00	-0.52
BLOK	cluster_10	Technology	0.50	-0.45
FXF	cluster_25	International Fixed Income	0.50	-0.36
HYMB	cluster_16	Municipal Bond	0.18	-0.18
UNG	cluster_19	Broad Commodities	0.07	-0.07
PRN	cluster_3	U.S. Large Cap Equity	0.09	-0.04
KIE	cluster_14	Financials	0.98	0.00
CQQQ	cluster_13	China Equity	0.09	0.01

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_19	11	FLAT	0.487	0.074	0.00016	
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_20	10	FLAT	0.303	0.094	0.00021	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.730	0.096	0.00024	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.647	0.206	0.00036	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.760	0.028	0.00007	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_19	11	FLAT	0.594	0.169	0.00039	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.734	0.119	0.00031	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.734	0.138	0.00039	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.694	0.165	0.00045	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.699	0.186	0.00046	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.715	0.180	0.00045	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_20	10	LONG	0.511	0.218	0.00056	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_5	58	LONG	0.847	0.125	0.00037	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.883	0.027	0.00001	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.883	0.053	0.00018	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.877	0.166	0.00032	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.819	0.203	0.00049	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.833	0.239	0.00049	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_20	10	FLAT	0.447	0.270	0.00074	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	FLAT	0.878	0.249	0.00047	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_20	10	FLAT	0.445	0.328	0.00083	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	LONG	0.780	0.276	0.00072	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.878	0.264	0.00055	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_12	17	LONG	0.743	0.356	0.00084	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_20	10	FLAT	0.499	0.351	0.00102	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.867	0.320	0.00071	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	LONG	0.853	0.256	0.00073	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_20	10	LONG	0.555	0.407	0.00107	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	FLAT	0.902	0.238	0.00065	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	FLAT	0.918	0.232	0.00059	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20